

Quotient maps of 2,3-uniform tilings of the plane on the torus

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Abstract

A 2-uniform tiling is an edge-to-edge tiling by regular polygons having 2 distinct transitivity classes of vertices. There are 20 distinct 2-uniform tilings (these are of 14 different types) on the plane, and since the plane is the universal cover of the torus, it is natural to explore maps on the torus that correspond to the 2-uniform tilings. In this article, we discuss that if a map is the quotient of a plane's 2-uniform lattice then what would be the bounds of the number of vertex orbits.

A 3-uniform tiling is an edge-to-edge tiling by regular polygons having 3 distinct transitivity classes of vertices. There are 61 distinct 3-uniform tilings on the plane. In this article, we discuss that if a map is the quotient of a plane's 3-uniform lattice then what would be the bounds of the number of vertex orbits.

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1 Introduction

A map is a connected 2-dimensional cell complex on a surface. Equivalently, it is a cellular embedding of a connected graph on a surface. For a map $\mathcal{K}$, let $V(\mathcal{K})$ be the vertex set of $\mathcal{K}$ and $u \in V(\mathcal{K})$. The faces containing u form a cycle (called the *face-cycle* at u) C_u in the dual graph of $\mathcal{K}$. That is, C_u is of the form $(F_{1,1} - \dots - F_{1,n_1}) - \dots - (F_{k,1} - \dots - F_{k,n_k}) - F_{1,1}$, where $F_{i,\ell}$ is a p_i -gon for $1 \leq \ell \leq n_i$, $1 \leq i \leq k$, $p_r \neq p_{r+1}$ for $1 \leq r \leq k-1$ and $p_k \neq p_1$. In this case, the vertex u is said to be of type $[p_1^{n_1}, \dots, p_k^{n_k}]$ (addition in the suffix is modulo k). A map $\mathcal{K}$ is said to be *k-semiequivelar* (or in short, *semiequivelar*) if $V(\mathcal{K}) = V_1 \sqcup V_2 \sqcup \dots \sqcup V_k$ such that the type of the vertices of V_i ($1 \leq i \leq k$) is same and the types of the vertices in V_i and V_j ($i \neq j$, $1 \leq i, j \leq k$) are different. In this article, we use semiequivelar to mean *k-semiequivelar* for some k . A semiequivelar map is said to be an *equivelar* map if it consists of single type of faces with all the vertices are of same type.

A *2-uniform tiling* is a tiling of the plane having 2 distinct transitivity classes of vertices. A vertex-transitive map (or tiling) is a map (or tiling) on a closed surface (or on the plane) on which the automorphism group acts transitively on the set of vertices. A 2-uniform tiling or map will have vertices that we could label X , and others that we could label Y . Each X vertex can be mapped onto every other X vertex, but cannot be mapped to any Y vertex.

A *semiregular* tiling of $\mathbb{R}^2$ is also known as *Archimedean*, or *homogeneous*, or *uniform* tiling. In [6], Grünbaum and Shephard showed that there are exactly eleven types of Archimedean tilings on the plane. These types are $[3^6]$, $[3^4, 6^1]$, $[3^3, 4^2]$, $[3^2, 4^1, 3^1, 4^1]$, $[3^1, 6^1, 3^1, 6^1]$, $[3^1, 4^1, 6^1, 4^1]$, $[3^1, 12^2]$, $[4^4]$, $[4^1, 6^1, 12^1]$, $[4^1, 8^2]$, $[6^3]$. Clearly, a *semiregular* tiling on $\mathbb{R}^2$ gives a semiequivelar map on $\mathbb{R}^2$. But, there are semiequivelar maps on the plane which are not (not isomorphic to) an Archimedean tiling. In fact, there exists $[p^q]$ equivelar maps on $\mathbb{R}^2$ whenever $1/p + 1/q < 1/2$ (see in [1], [5]). Thus, we have

Proposition 1.1. *There are infinitely many types of equivelar maps on the plane $\mathbb{R}^2$.*

We know that the plane is the universal cover of the torus. Since there are infinitely many equivelar maps on the plane, it is natural to ask what are the other types of semiequivelar maps that exist on the torus. Here we have the following result.

Proposition 1.2. *[2, 3] Let X be a semiequivelar map on a surface M . If M is the torus then the type of X is $[3^6]$, $[6^3]$, $[4^4]$, $[3^4, 6^1]$, $[3^3, 4^2]$, $[3^2, 4^1, 3^1, 4^1]$, $[3^1, 6^1, 3^1, 6^1]$, $[3^1, 4^1, 6^1, 4^1]$, $[3^1, 12^2]$, $[4^1, 8^2]$ or $[4^1, 6^1, 12^1]$.*

We know that all the Archimedean tilings are vertex-transitive. But, it is not true on the torus. Here, we know the following.

Proposition 1.3. *[2, 3] Let X be an equivelar map on the torus. If the type of X is $[3^6]$, $[4^4]$, $[6^3]$ or $[3^3, 4^2]$ then X is vertex-transitive. If the type is $[3^2, 4^1, 3^1, 4^1]$, $[3^1, 6^1, 3^1, 6^1]$, $[3^1, 4^1, 6^1, 4^1]$, $[3^1, 12^2]$, $[4^1, 8^2]$, $[3^4, 6^1]$ or $[4^1, 6^1, 12^1]$, then there exists a semiequivelar toroidal map which is not vertex-transitive.*

The 2-uniform tilings of the plane $\mathbb{R}^2$ are the generalization of vertex-transitive tilings on the plane. We know from [6, 7, 4] that there are twenty 2-uniform tiling of types

$$\begin{aligned} &[3^6; 3^3, 4^2], [3^6; 3^2, 4^1, 3^1, 4^1], [3^4, 6^1; 3^2, 6^2], [3^3, 4^2; 3^1, 4^1, 6^1, 4^1], [3^3, 4^2; 3^2, 4^1, 3^1, 4^1], \\ &[3^6; 3^2, 4^1, 12^1], [3^1, 4^1, 6^1, 4^1; 4^1, 6^1, 12^1], [3^2, 4^1, 3^1, 4^1; 3^1, 4^1, 6^1, 4^1], [3^2, 6^2; 3^1, 6^1, 3^1, 6^1], \\ &[3^1, 4^1, 3^1, 12^1; 3^1, 12^2], [3^1, 4^2, 6^1; 3^1, 4^1, 6^1, 4^1], [3^1, 4^2, 6^1; 3^1, 6^1, 3^1, 6^1], [3^3, 4^2; 4^4], [3^6; 3^4, 6^1]. \end{aligned}$$

on the plane (see in Section 3). We know that the plane is the universal cover of the torus. We also know that the maps of the above fourteen types also exist on the torus. Thus, it is natural to ask that if X is a map on the torus and $\eta: Y \rightarrow X$ is a covering map where Y is an 2-uniform tiling on the plane, then what would be the number of orbits of vertices of X . Here, we know the following.

Proposition 1.4. *[8] Let Z be a semiequivelar map on the torus and $\eta: K_5 \rightarrow Z$ be a covering map (K_5 given in Section 3). Let the vertices of Z form m $\text{Aut}(Z)$ -orbits. Then, $m \leq 4$.*

In this article, we prove the following.

Theorem 1.5. *Let X be a semiequivelar map on the torus and $\eta: Y \rightarrow X$ be a covering map. Let the vertices of X form m $\text{Aut}(X)$ -orbits. Let K_i for $1 \leq i \leq 20$ (in Section 3) denote the 2-uniform tilings on the plane.*

- (1) *If $Y = K_1, K_9, K_{11}, K_{14}$, or K_{17} then $m \leq 6$.*
- (2) *If $Y = K_2, K_7, K_{10}$, or K_{16} then $m \leq 4$.*
- (3) *If $Y = K_3, K_4, K_8, K_{12}, K_{13}$, or K_{15} then $m = 2$.*
- (4) *If $Y = K_{18}$ or K_{19} then $m \leq 3$.*
- (5) *If $Y = K_6$ then $m \leq 7$.*
- (6) *If $Y = K_{20}$ then $m \leq 9$.*

A *3-uniform tiling* is an edge-to-edge tiling by regular polygons having 3 distinct transitivity classes of vertices. A vertex-transitive map is a map on a closed surface on which the automorphism group acts transitively on the set of vertices. A 3-uniform tiling or map will have vertices that we could label X_1, X_2, X_3 . Each X_i vertex can be mapped onto every other X_i vertex, but cannot be mapped to any X_j for $j \neq i$ vertices.

The *k-uniform* tilings of the plane $\mathbb{R}^2$ are the generalization of vertex-transitive tilings on the plane. We know that the plane is the universal cover of the torus.

If $k = 3$, we know from [6, 7, 4] that there are 61 3-uniform tiling on the plane. In this article, we prove the following. If X is a map on the torus and $\eta: Y \rightarrow X$ is a covering map where Y is an 3-uniform tiling on the plane, then what would be the number of orbits of vertices of X . Precisely, we have the following result.

Theorem 1.6. *Let X be a semiequivelar map on the torus and $\eta: Y \rightarrow X$ be a covering map. Let the vertices of X form m $\text{Aut}(X)$ -orbits. Let K_i for $1 \leq i \leq 61$ (in Section 4) denote the 3-uniform tilings on the plane. If $X = K_1, K_2, \dots$, or, K_{61} then $m \leq 15$.*

2 Proof of Theorems 1.5

Let K_i (given in Section 3) be of type A , where

$$\begin{aligned} A \in \{ & [3^6; 3^4, 6^1], [3^6; 3^3, 4^2], [3^6; 3^2, 4^1, 3^1, 4^1], [3^6; 3^2, 4^1, 12^1], [3^6; 3^2, 6^2], \\ & [3^4, 6^1; 3^2, 6^2], [3^3, 4^2; 3^2, 4^1, 3^1, 4^1], [3^3, 4^2; 3^1, 4^1, 6^1, 4^1], [3^3, 4^2; 4^4], \\ & [3^2, 4^1, 3^1, 4^1; 3^1, 4^1, 6^1, 4^1], [3^2, 6^2; 3^1, 6^1, 3^1, 6^1], [3^1, 4^1, 3^1, 12^1; 3^1, 12^2], \\ & [3^1, 4^2, 6^1; 3^1, 4^1, 6^1, 4^1], [3^1, 4^2, 6^1; 3^1, 6^1, 3^1, 6^1], [3^1, 4^1, 6^1, 4^1; 4^1, 6^1, 12^1] \}. \end{aligned}$$

Grünbaum and G. C. Shephard [6, 7] and Krötenheerdt [4] have discussed the existence and uniqueness of the 2-uniform tilings K_i , $i = 1, 2, \dots, 20$ of the plane. Thus, we have the following.

Proposition 2.1. *The 2-uniform maps K_i ($1 \leq i \leq 20$) are unique up to isomorphism.*

Remark 2.2. *Let $S = \{0, 1, 2, \dots, m-1\}$. Let $S+n := \{0+n, 1+n, 2+n, \dots, m+n-1\}$. Define $S_m = \{S + mk : k \in \mathbb{Z} \text{ \& } 2 \mid (k-1)\}$. Let K_m be a map on the plane defined as follows. Let the vertex set of K_m be $\{(i, j) \in \mathbb{Z} \times \mathbb{Z}\}$ on $\mathbb{R}^2$. The points (i, j) and (i_1, j_1) are connected by an edge if $(i, j) \sim (i_1, j_1)$. Thus, the edges of K_m :*

- (1) $(i, j) \sim (i_1, j_1)$ if $|i - i_1| = 1$ and $j = j_1$, or $i = i_1$ and $|j - j_1| = 1$,
- (2) $(i, j) \sim (i+1, j+1), (t, j) \sim (t+1, j-1)$ if $i \in S_m, j$ is even and $t \in \mathbb{Z} \setminus S_m$.

Thus, we get an object K_m for each $m \geq 2$. Clearly, $K_x \not\cong K_y$ for $x \neq y$ since the number of continuous quadrilaterals are different. Therefore, there are infinitely many 2-semiequivelar tilings of the plane.

Proof of Theorem 1.5. Let X_1 be a semiequivelar map on the torus that is the quotient of the plane's 2-uniform lattice K_1 . Let the vertices of X_1 form m_1 $\text{Aut}(X_1)$ -orbits. Let K_1 be as in Section 3. Let $V_1 = V(K_1)$ be the vertex set of K_1 . Let H_1 be the group of all the translations of K_1 . So, $H_1 \leq \text{Aut}(K_1)$.

Since X_1 is a semiequivelar map on the torus that is the quotient of the plane's 2-uniform lattice K_1 (as by Proposition 2.1 K_1 is unique), so, we can assume, there is a polyhedral covering map $\eta_1: K_1 \rightarrow X_1$ where $X_1 = K_1/\Gamma_1$ for some fixed element (vertex, edge or face)

free subgroup $\Gamma_1 \leq \text{Aut}(K_1)$. Hence Γ_1 consists of translations and glide reflections. Since $X_1 = K_1/\Gamma_1$ is orientable, Γ_1 does not contain any glide reflection. Thus $\Gamma_1 \leq H_1$.

We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0, 0)$ of K_1 . Let $\alpha_1 := a_9 - a_0$, $\beta_1 := a_{14} - a_0$ and $\gamma_1 := a_{20} - a_0$ in K_1 . Then

$$H_1 := \langle x \mapsto x + \alpha_1, x \mapsto x + \beta_1, x \mapsto x + \gamma_1 \rangle.$$

Under the action of H_1 , vertices of K_1 form twelve orbits. Consider the subgroup G_1 of $\text{Aut}(K_1)$ generated by H_1 and the map (the half rotation) $x \mapsto -x$. So,

$$G_1 = \{ \alpha : x \mapsto \varepsilon x + m\alpha_1 + n\beta_1 + r\gamma_1 : \varepsilon = \pm 1, m, n, r \in \mathbb{Z} \} \cong H_1 \rtimes \mathbb{Z}_2.$$

Clearly, under the action of G_1 , vertices of K_1 form six orbits. The orbits are

$$O_1 := \langle a_0 \rangle, O_2 := \langle a_1 \rangle, O_3 := \langle a_2 \rangle, O_4 := \langle b_{18} \rangle, O_5 := \langle b_{20} \rangle, O_6 := \langle b_{28} \rangle.$$

Claim 1. If $K \leq H_1$ then $K \trianglelefteq G_1$. The alphabet K has already been used in the statement of 1.5

Let $g \in G_1$ and $k \in K$. Then $g(x) = \varepsilon x + ma + nb + rc$ and $k(x) = x + pa + qb + \ell c$ for some $m, n, r, p, q, \ell \in \mathbb{Z}$ and $\varepsilon \in \{1, -1\}$. Therefore,

$$\begin{aligned} (g \circ k \circ g^{-1})(x) &= (g \circ k)(\varepsilon(x - ma - nb - rc)) \\ &= g(\varepsilon(x - ma - nb - rc) + pa + qb + \ell c) \\ &= x - ma - nb - rc + \varepsilon(pa + qb + \ell c) + ma + nb + rc \\ &= x + \varepsilon(pa + qb + \ell c) \\ &= k^\varepsilon(x). \end{aligned}$$

Thus, $g \circ k \circ g^{-1} = k^\varepsilon \in K$. This completes the claim.

By Claim 1, Γ_1 is a normal subgroup of G_1 . Therefore, G_1/Γ_1 acts on $X_1 = K_1/\Gamma_1$. Since

$$O_1 := \langle a_0 \rangle, O_2 := \langle a_1 \rangle, O_3 := \langle a_2 \rangle, O_4 := \langle b_{18} \rangle, O_5 := \langle b_{20} \rangle, O_6 := \langle b_{28} \rangle.$$

are the G_1 -orbits, it follows that $\eta_1(O_j)$ for $j = 1, 2, \dots, 6$ are the (G_1/Γ_1) -orbits. Since the vertex set of K_1 is $\sqcup_{j=1}^6 \eta_j(O_j)$ and $G_1/\Gamma_1 \leq \text{Aut}(X_1)$, it follows that the number of $\text{Aut}(X_1)$ -orbits of vertices is ≤ 6 .

Let $X_9 = K_9/\Gamma_9$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_9 \leq \text{Aut}(K_9)$. Let the vertices of X_9 form m_9 $\text{Aut}(X_9)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_5 as the origin $(0, 0)$ of K_9 (see Section 3). Let $\alpha_9 := a_1 - a_0$, $\beta_9 := a_2 - a_0$ and $\gamma_9 := a_3 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_9 := \langle x \mapsto x + \alpha_9, x \mapsto x + \beta_9, x \mapsto x + \gamma_9 \rangle$ and

$$G_9 = \{ \alpha : x \mapsto \varepsilon x + m_9\alpha_9 + n_9\beta_9 + r_9\gamma_9 : \varepsilon = \pm 1, m_9, n_9, r_9 \in \mathbb{Z} \} \cong H_9 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_9 \trianglelefteq G_9$ and the number of G_9/Γ_9 -orbits of vertices of X_9 is six. Therefore, G_9/Γ_9 acts on $X_9 = K_9/\Gamma_9$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_{33} \rangle$, $O_5 = \langle a_8 \rangle$, $O_6 = \langle b_{34} \rangle$ are the G_9 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_{33} \rangle$, $O_5 = \langle a_8 \rangle$, $O_6 = \langle b_{34} \rangle$ are the (G_9/Γ_9) -orbits. Since the vertex set of X_9 is $\sqcup_{j=1}^6 \eta_j(O_j)$ and $G_9/\Gamma_9 \leq \text{Aut}(X_9)$, it follows that the number of $\text{Aut}(X_9)$ -orbits of vertices is ≤ 6 .

Let $X_{11} = K_{11}/\Gamma_{11}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{11} \leq \text{Aut}(K_{11})$. Let the vertices of X_{11} form m_{11} $\text{Aut}(X_{11})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{11} . Let $\alpha_{11} := a_{15} - a_0$, $\beta_{11} := a_6 - a_0$ and $\gamma_{11} := a_{37} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{11} := \langle x \mapsto x + \alpha_{11}, x \mapsto x + \beta_{11}, x \mapsto x + \gamma_{11} \rangle$ and

$$G_{11} = \{\alpha : x \mapsto \varepsilon x + m_{11}\alpha_{11} + n_{11}\beta_{11} + r_{11}\gamma_{11} : \varepsilon = \pm 1, m_{11}, n_{11}, r_{11} \in \mathbb{Z}\} \cong H_{11} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{11} \trianglelefteq G_{11}$ and the number of G_{11}/Γ_{11} -orbits of vertices of X_{11} is six. Therefore, G_{11}/Γ_{11} acts on $X_{11} = K_{11}/\Gamma_{11}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_2 \rangle$, $O_6 = \langle b_{46} \rangle$ are the G_{11} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_2 \rangle$, $O_6 = \langle b_{46} \rangle$ are the (G_{11}/Γ_{11}) -orbits. Since the vertex set of X_{11} is $\sqcup_{j=1}^6 \eta_{11}(O_j)$ and $G_{11}/\Gamma_{11} \leq \text{Aut}(X_{11})$, it follows that the number of $\text{Aut}(X_{11})$ -orbits of vertices is ≤ 6 .

Let $X_{14} = K_{14}/\Gamma_{14}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{14} \leq \text{Aut}(K_{14})$. Let the vertices of X_{14} form m_{14} $\text{Aut}(X_{14})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{14} . Let $\alpha_{14} := a_9 - a_3$, $\beta_{14} := a_{30} - a_3$ and $\gamma_{14} := a_{34} - a_3 \in \mathbb{R}^2$. Similarly as above, define $H_{14} := \langle x \mapsto x + \alpha_{14}, x \mapsto x + \beta_{14}, x \mapsto x + \gamma_{14} \rangle$ and

$$G_{14} = \{\alpha : x \mapsto \varepsilon x + m_{14}\alpha_{14} + n_{14}\beta_{14} + r_{14}\gamma_{14} : \varepsilon = \pm 1, m_{14}, n_{14}, r_{14} \in \mathbb{Z}\} \cong H_{14} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{14} \trianglelefteq G_{14}$ and the number of G_{14}/Γ_{14} -orbits of vertices of X_{14} is six. Therefore, G_{14}/Γ_{14} acts on $X_{14} = K_{14}/\Gamma_{14}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_2 \rangle$, $O_6 = \langle b_4 \rangle$ are the G_{14} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_2 \rangle$, $O_6 = \langle b_4 \rangle$ are the (G_{14}/Γ_{14}) -orbits. Since the vertex set of X_{14} is $\sqcup_{j=1}^6 \eta_{14}(O_j)$ and $G_{14}/\Gamma_{14} \leq \text{Aut}(X_{14})$, it follows that the number of $\text{Aut}(X_{14})$ -orbits of vertices is ≤ 6 .

Let $X_{17} = K_{17}/\Gamma_{17}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{17} \leq \text{Aut}(K_{17})$. Let the vertices of X_{17} form m_{17} $\text{Aut}(X_{17})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{17} . Let $\alpha_{17} := a_{20} - a_4$, $\beta_{17} := a_7 - a_4$ and $\gamma_{17} := a_{16} - a_4 \in \mathbb{R}^2$. Similarly as above, define $H_{17} := \langle x \mapsto x + \alpha_{17}, x \mapsto x + \beta_{17}, x \mapsto x + \gamma_{17} \rangle$ and

$$G_{17} = \{\alpha : x \mapsto \varepsilon x + m_{17}\alpha_{17} + n_{17}\beta_{17} + r_{17}\gamma_{17} : \varepsilon = \pm 1, m_{17}, n_{17}, r_{17} \in \mathbb{Z}\} \cong H_{17} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{17} \trianglelefteq G_{17}$ and the number of G_{17}/Γ_{17} -orbits of vertices of X_{17} is nine. Therefore, G_{17}/Γ_{17} acts on $X_{17} = K_{17}/\Gamma_{17}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_1 \rangle$, $O_6 = \langle b_2 \rangle$ are the G_{17} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_1 \rangle$, $O_6 = \langle b_2 \rangle$ are the (G_{17}/Γ_{17}) -orbits. Since the vertex set of X_{17} is $\sqcup_{j=1}^6 \eta_{17}(O_j)$ and $G_{17}/\Gamma_{17} \leq \text{Aut}(X_{17})$, it follows that the number of $\text{Aut}(X_{17})$ -orbits of vertices is ≤ 6 . Thus, it completes the part (1).

Let $X_2 = K_2/\Gamma_2$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_2 \leq \text{Aut}(K_2)$. Let the vertices of X_2 form m_2 $\text{Aut}(X_2)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_2 . Let $\alpha_2 := a_6 - a_0$, $\beta_2 := a_{18} - a_0$ and $\gamma_2 := a_{24} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_2 := \langle x \mapsto x + \alpha_2, x \mapsto x + \beta_2, x \mapsto x + \gamma_2 \rangle$ and

$$G_2 = \{\alpha : x \mapsto \varepsilon x + m_1\alpha_2 + n_1\beta_2 + r_1\gamma_2 : \varepsilon = \pm 1, m_1, n_1, r_1 \in \mathbb{Z}\} \cong H_2 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_2 \trianglelefteq G_2$ and the number of G_2/Γ_2 -orbits of vertices of X_2 is four. Therefore, G_2/Γ_2 acts on $X_2 = K_2/\Gamma_2$. Since

$$O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle a_2 \rangle, O_4 = \langle b_0 \rangle$$

are the G_2 -orbits, it follows that $\eta_2(O_j)$ for $j = 1, 2, 3, 4$ are the (G_2/Γ_2) -orbits. Since the vertex set of Y is $\sqcup_{j=1}^4 \eta_j(O_j)$ and $G_2/\Gamma_2 \leq \text{Aut}(X_2)$, it follows that the number of $\text{Aut}(X_2)$ -orbits of vertices is ≤ 4 .

Let $X_7 = K_7/\Gamma_7$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_7 \leq \text{Aut}(K_7)$. Let the vertices of X_7 form m_7 $\text{Aut}(X_7)$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_7 as the origin $(0, 0)$ of K_7 . Let $\alpha_7 := b_{22} - b_0$, $\beta_7 := b_{28} - b_0$ and $\gamma_7 := b_{41} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_7 := \langle x \mapsto x + \alpha_7, x \mapsto x + \beta_7, x \mapsto x + \gamma_7 \rangle$ and

$$G_7 = \{ \alpha : x \mapsto \varepsilon x + m_7 \alpha_7 + n_7 \beta_7 + r_7 \gamma_7 : \varepsilon = \pm 1, m_7, n_7, r_7 \in \mathbb{Z} \} \cong H_7 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_7 \trianglelefteq G_7$ and the number of G_7/Γ_7 -orbits of vertices of X_7 is four. Therefore, G_7/Γ_7 acts on $X_7 = K_7/\Gamma_7$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_5 \rangle$, $O_4 = \langle b_6 \rangle$ are the G_7 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_5 \rangle$, $O_4 = \langle b_6 \rangle$ are the (G_7/Γ_7) -orbits. Since the vertex set of X_7 is $\sqcup_{j=1}^4 \eta_j(O_j)$ and $G_7/\Gamma_7 \leq \text{Aut}(X_7)$, it follows that the number of $\text{Aut}(X_7)$ -orbits of vertices is ≤ 4 .

Let $X_{10} = K_{10}/\Gamma_{10}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{10} \leq \text{Aut}(K_{10})$. Let the vertices of X_{10} form m_{10} $\text{Aut}(X_{10})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_1 as the origin $(0, 0)$ of K_{10} . Let $\alpha_{10} := b_6 - b_0$ and $\beta_{10} := b_{23} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{10} := \langle x \mapsto x + \alpha_{10}, x \mapsto x + \beta_{10} \rangle$ and

$$G_{10} = \{ \alpha : x \mapsto \varepsilon x + m_{10} \alpha_{10} + n_{10} \beta_{10} : \varepsilon = \pm 1, m_{10}, n_{10} \in \mathbb{Z} \} \cong H_{10} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{10} \trianglelefteq G_{10}$ and the number of G_{10}/Γ_{10} -orbits of vertices of X_{10} is three. Therefore, G_{10}/Γ_{10} acts on $X_{10} = K_{10}/\Gamma_{10}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_2 \rangle$ are the G_{10} -orbits, it follows that $\eta_{10}(O_1), \eta_{10}(O_2), \eta_{10}(O_3), \eta_{10}(O_4)$ are the (G_{10}/Γ_{10}) -orbits. Since the vertex set of X_{10} is $\eta_{10}(O_1) \sqcup \eta_{10}(O_2) \sqcup \eta_{10}(O_3) \sqcup \eta_{10}(O_4)$ and $G_{10}/\Gamma_{10} \leq \text{Aut}(X_{10})$, it follows that the number of $\text{Aut}(X_{10})$ -orbits of vertices is ≤ 4 .

Let $X_{16} = K_{16}/\Gamma_{16}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{16} \leq \text{Aut}(K_{16})$. Let the vertices of X_{16} form m_{16} $\text{Aut}(X_{16})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_4 as the origin $(0, 0)$ of K_{16} . Let $\alpha_{16} := a_4 - a_1$ and $\beta_{16} := a_8 - a_1 \in \mathbb{R}^2$. Similarly as above, define $H_{16} := \langle x \mapsto x + \alpha_{16}, x \mapsto x + \beta_{16} \rangle$ and

$$G_{16} = \{ \alpha : x \mapsto \varepsilon x + m_{16} \alpha_{16} + n_{16} \beta_{16} : \varepsilon = \pm 1, m_{16}, n_{16} \in \mathbb{Z} \} \cong H_{16} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{16} \trianglelefteq G_{16}$ and the number of G_{16}/Γ_{16} -orbits of vertices of X_{16} is three. Therefore, G_{16}/Γ_{16} acts on $X_{16} = K_{16}/\Gamma_{16}$. Since $O_1 = \langle a_1 \rangle$, $O_2 = \langle b_1 \rangle$, $O_3 = \langle a_3 \rangle$, $O_4 = \langle b_2 \rangle$ are the G_{16} -orbits, it follows that $\eta_{16}(O_1), \eta_{16}(O_2), \eta_{16}(O_3), \eta_{16}(O_4)$ are the (G_{16}/Γ_{16}) -orbits. Since the vertex set of X_{16} is $\eta_{16}(O_1) \sqcup \eta_{16}(O_2) \sqcup \eta_{16}(O_3) \sqcup \eta_{16}(O_4)$ and $G_{16}/\Gamma_{16} \leq \text{Aut}(X_{16})$, it follows that the number of $\text{Aut}(X_{16})$ -orbits of vertices is ≤ 2 and hence, $m_{16} \leq 4$. Thus, it completes the part (2).

Let $X_3 = K_3/\Gamma_3$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_3 \leq \text{Aut}(K_3)$. Let the vertices of X_3 form m_3 $\text{Aut}(X_3)$ -orbits.

We take the middle point of the line segment joining vertices a_0 and a_{11} as the origin $(0, 0)$ of K_3 . Let $\alpha_3 := a_1 - a_0$ and $\beta_3 := a_{35} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_3 := \langle x \mapsto x + \alpha_3, x \mapsto x + \beta_3 \rangle$ and

$$G_3 = \{\alpha : x \mapsto \varepsilon x + m_3 \alpha_3 + n_3 \beta_3 : \varepsilon = \pm 1, m_3, n_3 \in \mathbb{Z}\} \cong H_3 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_3 \trianglelefteq G_3$ and the number of G_3/Γ_3 -orbits of vertices of X_3 is two. Therefore, G_3/Γ_3 acts on $X_3 = K_3/\Gamma_3$. Since $O_1 = \langle a_1 \rangle, O_2 = \langle b_1 \rangle$ are the G_3 -orbits, it follows that $\eta_3(O_1), \eta_3(O_2)$ are the (G_3/Γ_3) -orbits. Since the vertex set of X_3 is $\eta_3(O_1) \sqcup \eta_3(O_2)$ and $G_3/\Gamma_3 \leq \text{Aut}(X_3)$, it follows that the number of $\text{Aut}(X_3)$ -orbits of vertices is ≤ 2 and hence, $m_3 = 2$.

Let $X_4 = K_4/\Gamma_4$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_4 \leq \text{Aut}(K_4)$. Let the vertices of X_4 form m_4 $\text{Aut}(X_4)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{12} as the origin $(0, 0)$ of K_4 . Let $\alpha_4 := a_1 - a_0$ and $\beta_4 := a_{43} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_4 := \langle x \mapsto x + \alpha_4, x \mapsto x + \beta_4 \rangle$ and

$$G_4 = \{\alpha : x \mapsto \varepsilon x + m_4 \alpha_4 + n_4 \beta_4 : \varepsilon = \pm 1, m_4, n_4 \in \mathbb{Z}\} \cong H_4 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_4 \trianglelefteq G_4$ and the number of G_4/Γ_4 -orbits of vertices of X_4 is two. Therefore, G_4/Γ_4 acts on $X_4 = K_4/\Gamma_4$. Since $O_1 = \langle a_1 \rangle, O_2 = \langle b_1 \rangle$ are the G_4 -orbits, it follows that $\eta_4(O_1), \eta_4(O_2)$ are the (G_4/Γ_4) -orbits. Since the vertex set of X_4 is $\eta_4(O_1) \sqcup \eta_4(O_2)$ and $G_4/\Gamma_4 \leq \text{Aut}(X_4)$, it follows that the number of $\text{Aut}(X_4)$ -orbits of vertices is ≤ 2 and hence, $m_4 = 2$.

Let $X_8 = K_8/\Gamma_8$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_8 \leq \text{Aut}(K_8)$. Let the vertices of X_8 form m_8 $\text{Aut}(X_8)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_1 as the origin $(0, 0)$ of K_8 . Let $\alpha_8 := a_3 - a_0$ and $\beta_8 := a_4 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_8 := \langle x \mapsto x + \alpha_8, x \mapsto x + \beta_8 \rangle$ and

$$G_8 = \{\alpha : x \mapsto \varepsilon x + m_8 \alpha_8 + n_8 \beta_8 : \varepsilon = \pm 1, m_8, n_8 \in \mathbb{Z}\} \cong H_8 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_8 \trianglelefteq G_8$ and the number of G_8/Γ_8 -orbits of vertices of X_8 is two. Therefore, G_8/Γ_8 acts on $X_8 = K_8/\Gamma_8$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle b_0 \rangle$ are the G_8 -orbits, it follows that $\eta_8(O_1), \eta_8(O_2)$ are the (G_8/Γ_8) -orbits. Since the vertex set of X_8 is $\eta_8(O_1) \sqcup \eta_8(O_2)$ and $G_8/\Gamma_8 \leq \text{Aut}(X_8)$, it follows that the number of $\text{Aut}(X_8)$ -orbits of vertices is ≤ 2 and hence, $m_8 = 2$.

Let $X_{12} = K_{12}/\Gamma_{12}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{12} \leq \text{Aut}(K_{12})$. Let the vertices of X_{12} form m_{12} $\text{Aut}(X_{12})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_1 as the origin $(0, 0)$ of K_{12} . Let $\alpha_{12} := a_1 - a_0$ and $\beta_{12} := a_{12} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{12} := \langle x \mapsto x + \alpha_{12}, x \mapsto x + \beta_{12} \rangle$ and

$$G_{12} = \{\alpha : x \mapsto \varepsilon x + m_{12} \alpha_{12} + n_{12} \beta_{12} : \varepsilon = \pm 1, m_{12}, n_{12} \in \mathbb{Z}\} \cong H_{12} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{12} \trianglelefteq G_{12}$ and the number of G_{12}/Γ_{12} -orbits of vertices of X_{12} is two. Therefore, G_{12}/Γ_{12} acts on $X_{12} = K_{12}/\Gamma_{12}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle b_0 \rangle$ are the G_{12} -orbits, it follows that $\eta_{12}(O_1), \eta_{12}(O_2)$ are the (G_{12}/Γ_{12}) -orbits. Since the vertex set of X_{12} is $\eta_{12}(O_1) \sqcup \eta_{12}(O_2)$ and $G_{12}/\Gamma_{12} \leq \text{Aut}(X_{12})$, it follows that the number of $\text{Aut}(X_{12})$ -orbits of vertices is ≤ 2 and hence, $m_{12} = 2$.

Let $X_{13} = K_{13}/\Gamma_{13}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{13} \leq \text{Aut}(K_{13})$. Let the vertices of X_{13} form m_{13} $\text{Aut}(X_{13})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{12} as the origin $(0,0)$ of K_{13} . Let $\alpha_{13} := a_1 - a_0$ and $\beta_{13} := a_{19} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{13} := \langle x \mapsto x + \alpha_{13}, x \mapsto x + \beta_{13} \rangle$ and

$$G_{13} = \{ \alpha : x \mapsto \varepsilon x + m_{13}\alpha_{13} + n_{13}\beta_{13} : \varepsilon = \pm 1, m_{13}, n_{13} \in \mathbb{Z} \} \cong H_{13} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{13} \trianglelefteq G_{13}$ and the number of G_{13}/Γ_{13} -orbits of vertices of X_{13} is two. Therefore, G_{13}/Γ_{13} acts on $X_{13} = K_{13}/\Gamma_{13}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle b_0 \rangle$ are the G_{13} -orbits, it follows that $\eta_{13}(O_1), \eta_{13}(O_2)$ are the (G_{13}/Γ_{13}) -orbits. Since the vertex set of X_{13} is $\eta_{13}(O_1) \sqcup \eta_{13}(O_2)$ and $G_{13}/\Gamma_{13} \leq \text{Aut}(X_{13})$, it follows that the number of $\text{Aut}(X_{13})$ -orbits of vertices is ≤ 2 and hence, $m_{13} = 2$.

Let $X_{15} = K_{15}/\Gamma_{15}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{15} \leq \text{Aut}(K_{15})$. Let the vertices of X_{15} form m_{15} $\text{Aut}(X_{15})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{19} as the origin $(0,0)$ of K_{15} . Let $\alpha_{15} := a_2 - a_0$ and $\beta_{15} := a_{10} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{15} := \langle x \mapsto x + \alpha_{15}, x \mapsto x + \beta_{15} \rangle$ and

$$G_{15} = \{ \alpha : x \mapsto \varepsilon x + m_{15}\alpha_{15} + n_{15}\beta_{15} : \varepsilon = \pm 1, m_{15}, n_{15} \in \mathbb{Z} \} \cong H_{15} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{15} \trianglelefteq G_{15}$ and the number of G_{15}/Γ_{15} -orbits of vertices of X_{15} is two. Therefore, G_{15}/Γ_{15} acts on $X_{15} = K_{15}/\Gamma_{15}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle b_0 \rangle$ are the G_{15} -orbits, it follows that $\eta_{15}(O_1), \eta_{15}(O_2)$ are the (G_{15}/Γ_{15}) -orbits. Since the vertex set of X_{15} is $\eta_{15}(O_1) \sqcup \eta_{15}(O_2)$ and $G_{15}/\Gamma_{15} \leq \text{Aut}(X_{15})$, it follows that the number of $\text{Aut}(X_{15})$ -orbits of vertices is ≤ 2 and hence, $m_{15} = 2$. Thus, it completes the part (3).

Let $X_{18} = K_{18}/\Gamma_{18}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{18} \leq \text{Aut}(K_{18})$. Let the vertices of X_{18} form m_{18} $\text{Aut}(X_{18})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{28} as the origin $(0,0)$ of K_{18} . Let $\alpha_{18} := a_2 - a_0$ and $\beta_{18} := a_{20} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{18} := \langle x \mapsto x + \alpha_{18}, x \mapsto x + \beta_{18} \rangle$ and

$$G_{18} = \{ \alpha : x \mapsto \varepsilon x + m_{18}\alpha_{18} + n_{18}\beta_{18} : \varepsilon = \pm 1, m_{18}, n_{18} \in \mathbb{Z} \} \cong H_{18} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{18} \trianglelefteq G_{18}$ and the number of G_{18}/Γ_{18} -orbits of vertices of X_{18} is three. Therefore, G_{18}/Γ_{18} acts on $X_{18} = K_{18}/\Gamma_{18}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle b_1 \rangle$ are the G_{18} -orbits, it follows that $\eta_{18}(O_1), \eta_{18}(O_2), \eta_{18}(O_3)$ are the (G_{18}/Γ_{18}) -orbits. Since the vertex set of X_{18} is $\eta_{18}(O_1) \sqcup \eta_{18}(O_2) \sqcup \eta_{18}(O_3)$ and $G_{18}/\Gamma_{18} \leq \text{Aut}(X_{18})$, it follows that the number of $\text{Aut}(X_{18})$ -orbits of vertices is ≤ 3 .

Let $X_{19} = K_{19}/\Gamma_{19}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{19} \leq \text{Aut}(K_{19})$. Let the vertices of X_{19} form m_{19} $\text{Aut}(X_{19})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_2 as the origin $(0,0)$ of K_{19} . Let $\alpha_{19} := b_0 - b_2$ and $\beta_{19} := b_7 - b_2 \in \mathbb{R}^2$. Similarly as above, define $H_{19} := \langle x \mapsto x + \alpha_{19}, x \mapsto x + \beta_{19} \rangle$ and

$$G_{19} = \{ \alpha : x \mapsto \varepsilon x + m_{19}\alpha_{19} + n_{19}\beta_{19} : \varepsilon = \pm 1, m_{19}, n_{19} \in \mathbb{Z} \} \cong H_{19} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{19} \trianglelefteq G_{19}$ and the number of G_{19}/Γ_{19} -orbits of vertices of X_{19} is three. Therefore, G_{19}/Γ_{19} acts on $X_{19} = K_{19}/\Gamma_{19}$. Since $O_1 = \langle b_0 \rangle,$

$O_2 = \langle a_2 \rangle$, $O_3 = \langle a_3 \rangle$ are the G_{19} -orbits, it follows that $\eta_{19}(O_1), \eta_{19}(O_2), \eta_{19}(O_3)$ are the (G_{19}/Γ_{19}) -orbits. Since the vertex set of X_{19} is $\eta_{19}(O_1) \sqcup \eta_{19}(O_2) \sqcup \eta_{19}(O_3)$ and $G_{19}/\Gamma_{19} \leq \text{Aut}(X_{19})$, it follows that the number of $\text{Aut}(X_{19})$ -orbits of vertices is ≤ 3 . Thus, it completes the part (4).

Let $X_6 = K_6/\Gamma_6$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_6 \leq \text{Aut}(K_6)$. Let the vertices of X_6 form m_6 $\text{Aut}(X_6)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_6 as the origin $(0, 0)$ of K_6 . Let $\alpha_6 := a_{23} - a_3$, $\beta_6 := a_{16} - a_6$ and $\gamma_6 := a_{39} - a_7 \in \mathbb{R}^2$. Similarly as above, define $H_6 := \langle x \mapsto x + \alpha_6, x \mapsto x + \beta_6, x \mapsto x + \gamma_6 \rangle$ and

$$G_6 = \{ \alpha : x \mapsto \varepsilon x + m_6 \alpha_6 + n_6 \beta_6 + r_6 \gamma_6 : \varepsilon = \pm 1, m_6, n_6, r_6 \in \mathbb{Z} \} \cong H_6 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_6 \trianglelefteq G_6$ and the number of G_6/Γ_6 -orbits of vertices of X_6 is seven. Therefore, G_6/Γ_6 acts on $X_6 = K_6/\Gamma_6$. Since $O_j = \langle a_j \rangle$, $j = 0, 1, \dots, 5$ $O_6 = \langle b_1 \rangle$ are the G_6 -orbits, it follows that $\eta_6(O_j)$, $j = 0, 1, \dots, 6$ are the (G_6/Γ_6) -orbits. Since the vertex set of X_6 is $\sqcup_{j=0}^6 \eta_6(O_j)$ and $G_6/\Gamma_6 \leq \text{Aut}(X_6)$, it follows that the number of $\text{Aut}(X_6)$ -orbits of vertices is ≤ 7 . Thus, it completes the part (5).

Let $X_{20} = K_{20}/\Gamma_{20}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{20} \leq \text{Aut}(K_{20})$. Let the vertices of X_{20} form m_{20} $\text{Aut}(X_{20})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_6 as the origin $(0, 0)$ of K_{20} . Let $\alpha_{20} := a_{24} - a_7$, $\beta_{20} := a_{14} - a_5$ and $\gamma_{20} := a_{20} - a_4 \in \mathbb{R}^2$. Similarly as above, define $H_{20} := \langle x \mapsto x + \alpha_{20}, x \mapsto x + \beta_{20}, x \mapsto x + \gamma_{20} \rangle$ and

$$G_{20} = \{ \alpha : x \mapsto \varepsilon x + m_{20} \alpha_{20} + n_{20} \beta_{20} + r_{20} \gamma_{20} : \varepsilon = \pm 1, m_{20}, n_{20}, r_{20} \in \mathbb{Z} \} \cong H_{20} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{20} \trianglelefteq G_{20}$ and the number of G_{20}/Γ_{20} -orbits of vertices of X_{20} is nine. Therefore, G_{20}/Γ_{20} acts on $X_{20} = K_{20}/\Gamma_{20}$. Since $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_2 \rangle$, $O_3 = \langle a_3 \rangle$, $O_4 = \langle a_4 \rangle$, $O_5 = \langle a_5 \rangle$, $O_6 = \langle a_6 \rangle$, $O_7 = \langle b_3 \rangle$, $O_8 = \langle b_0 \rangle$, $O_9 = \langle b_{11} \rangle$ are the G_{20} -orbits, it follows that $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_2 \rangle$, $O_3 = \langle a_3 \rangle$, $O_4 = \langle a_4 \rangle$, $O_5 = \langle a_5 \rangle$, $O_6 = \langle a_6 \rangle$, $O_7 = \langle b_3 \rangle$, $O_8 = \langle b_0 \rangle$, $O_9 = \langle b_{11} \rangle$ are the (G_{20}/Γ_{20}) -orbits. Since the vertex set of X_{20} is $\sqcup_{j=1}^9 \eta_{20}(O_j)$ and $G_{20}/\Gamma_{20} \leq \text{Aut}(X_{20})$, it follows that the number of $\text{Aut}(X_{20})$ -orbits of vertices is ≤ 9 . Thus, it completes the part (6). $\square$

5 Proof of Theorem 1.6

Proposition 5.1. *The maps K_i ($1 \leq i \leq 61$) (in Section 4) are unique up to isomorphism.*

Proof of Theorem 1.6. Let X_1 be a semiequivelar map on the torus that is the quotient of the plane's 3-uniform lattice K_1 . Let the vertices of X_1 form m_1 $\text{Aut}(X_1)$ -orbits. Let K_1 be as in Section 4. Let $V_1 = V(K_1)$ be the vertex set of K_1 . Let H_1 be the group of all the translations of K_1 . So, $H_1 \leq \text{Aut}(K_1)$.

Since X_1 is a semiequivelar map on the torus that is the quotient of the plane's 3-uniform lattice K_1 (as by Proposition 5.1 K_1 is unique), so, we can assume, there is a polyhedral covering map $\eta_1 : K_1 \rightarrow X_1$ where $X_1 = K_1/\Gamma_1$ for some fixed element (vertex, edge or face) free subgroup $\Gamma_1 \leq \text{Aut}(K_1)$. Hence Γ_1 consists of translations and glide reflections. Since $X_1 = K_1/\Gamma_1$ is orientable, Γ_1 does not contain any glide reflection. Thus $\Gamma_1 \leq H_1$.

We take the middle point of the line segment joining vertices a_0 and a_1 as the origin $(0, 0)$ of K_1 . Let $\alpha_1 := a_2 - a_0$ and $\beta_1 := a_7 - a_0$ in K_1 . Then

$$H_1 := \langle x \mapsto x + \alpha_1, x \mapsto x + \beta_1 \rangle.$$

Under the action of H_1 , vertices of K_1 form five orbits. Consider the subgroup G_1 of $\text{Aut}(K_1)$ generated by H_1 and the map (the half rotation) $x \mapsto -x$. So,

$$G_1 = \{ \alpha : x \mapsto \varepsilon x + m\alpha_1 + n\beta_1 : \varepsilon = \pm 1, m, n \in \mathbb{Z} \} \cong H_1 \rtimes \mathbb{Z}_2.$$

Clearly, under the action of G_1 , vertices of K_1 form three orbits. The orbits are

$$O_1 := \langle a_0 \rangle, O_2 := \langle b_0 \rangle, O_3 := \langle c_0 \rangle.$$

Claim 1. If $S \leq H_1$ then $S \trianglelefteq G_1$.

Let $g \in G_1$ and $s \in S$. Then $g(x) = \varepsilon x + ma + nb + rc$ and $s(x) = x + pa + qb + \ell c$ for some $m, n, r, p, q, \ell \in \mathbb{Z}$ and $\varepsilon \in \{1, -1\}$. Therefore,

$$\begin{aligned} (g \circ s \circ g^{-1})(x) &= (g \circ s)(\varepsilon(x - ma - nb - rc)) \\ &= g(\varepsilon(x - ma - nb - rc) + pa + qb + \ell c) \\ &= x - ma - nb - rc + \varepsilon(pa + qb + \ell c) + ma + nb + rc \\ &= x + \varepsilon(pa + qb + \ell c) \\ &= s^\varepsilon(x). \end{aligned}$$

Thus, $g \circ s \circ g^{-1} = s^\varepsilon \in S$. This completes the claim.

By Claim 1, Γ_1 is a normal subgroup of G_1 . Therefore, G_1/Γ_1 acts on $X_1 = K_1/\Gamma_1$. Since

$$O_1 := \langle a_0 \rangle, O_2 := \langle b_0 \rangle, O_3 := \langle c_0 \rangle$$

are the G_1 -orbits, it follows that $\eta_1(O_j)$ for $j = 1, 2, \dots, 6$ are the (G_1/Γ_1) -orbits. Since the vertex set of K_1 is $\sqcup_{j=1}^3 \eta_j(O_j)$ and $G_1/\Gamma_1 \leq \text{Aut}(X_1)$, it follows that the number of $\text{Aut}(X_1)$ -orbits of vertices is 3.

Let $X_2 = K_2/\Gamma_2$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_2 \leq \text{Aut}(K_2)$. Let the vertices of X_2 form m_2 $\text{Aut}(X_2)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_1 as the origin $(0, 0)$ of K_2 (see Section 4). Let $\alpha_2 := a_4 - a_0$ and $\beta_2 := a_2 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_2 := \langle x \mapsto x + \alpha_2, x \mapsto x + \beta_2 \rangle$ and

$$G_2 = \{ \alpha : x \mapsto \varepsilon x + m_2\alpha_2 + n_2\beta_2 : \varepsilon = \pm 1, m_2, n_2 \in \mathbb{Z} \} \cong H_2 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_2 \trianglelefteq G_2$ and the number of G_2/Γ_2 -orbits of vertices of X_2 is three. Therefore, G_2/Γ_2 acts on $X_2 = K_2/\Gamma_2$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle c_0 \rangle$ are the G_9 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle c_0 \rangle$ are the (G_2/Γ_2) -orbits. Since the vertex set of X_2 is $\sqcup_{j=1}^3 \eta_2(O_j)$ and $G_2/\Gamma_2 \leq \text{Aut}(X_2)$, it follows that the number of $\text{Aut}(X_2)$ -orbits of vertices is 3.

Let $X_3 = K_3/\Gamma_3$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_3 \leq \text{Aut}(K_3)$. Let the vertices of X_3 form m_3 $\text{Aut}(X_3)$ -orbits. We take the middle point of the line segment joining vertices b_1 and c_1 as the origin $(0, 0)$ of K_3 (see Section 4). Let $\alpha_3 := c_{22} - c_1$ and $\beta_3 := c_5 - c_1 \in \mathbb{R}^2$. Similarly as above, define $H_3 := \langle x \mapsto x + \alpha_3, x \mapsto x + \beta_3, x \mapsto x + \gamma_3 \rangle$ and

$$G_3 = \{ \alpha : x \mapsto \varepsilon x + m_3 \alpha_3 + n_3 \beta_3 : \varepsilon = \pm 1, m_3, n_3 \in \mathbb{Z} \} \cong H_3 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_3 \trianglelefteq G_3$ and the number of G_3/Γ_3 -orbits of vertices of X_3 is eight. Therefore, G_3/Γ_3 acts on $X_3 = K_3/\Gamma_3$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_5 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_2 \rangle$, $O_6 = \langle c_0 \rangle$, $O_7 = \langle c_1 \rangle$, $O_8 = \langle c_2 \rangle$ are the G_9 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_5 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_2 \rangle$, $O_6 = \langle c_0 \rangle$, $O_7 = \langle c_1 \rangle$, $O_8 = \langle c_2 \rangle$ are the (G_3/Γ_3) -orbits. Since the vertex set of X_3 is $\sqcup_{j=1}^8 \eta_3(O_j)$ and $G_3/\Gamma_3 \leq \text{Aut}(X_3)$, it follows that the number of $\text{Aut}(X_3)$ -orbits of vertices is ≤ 8 .

Let $X_4 = K_4/\Gamma_4$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_4 \leq \text{Aut}(K_4)$. Let the vertices of X_4 form m_4 $\text{Aut}(X_4)$ -orbits. We take the middle point of the line segment joining vertices b_3 and b_4 as the origin $(0, 0)$ of K_4 (see Section 4). Let $\alpha_4 := b_4 - b_3$ and $\beta_4 := b_{13} - b_3 \in \mathbb{R}^2$. Similarly as above, define $H_4 := \langle x \mapsto x + \alpha_4, x \mapsto x + \beta_4 \rangle$ and

$$G_4 = \{ \alpha : x \mapsto \varepsilon x + m_4 \alpha_4 + n_4 \beta_4 : \varepsilon = \pm 1, m_4, n_4 \in \mathbb{Z} \} \cong H_4 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_4 \trianglelefteq G_4$ and the number of G_4/Γ_4 -orbits of vertices of X_4 is three. Therefore, G_4/Γ_4 acts on $X_4 = K_4/\Gamma_4$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_0 \rangle$ are the G_4 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_0 \rangle$ are the (G_4/Γ_4) -orbits. Since the vertex set of X_4 is $\sqcup_{j=1}^3 \eta_4(O_j)$ and $G_4/\Gamma_4 \leq \text{Aut}(X_4)$, it follows that the number of $\text{Aut}(X_4)$ -orbits of vertices is 3.

Let $X_5 = K_5/\Gamma_5$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_5 \leq \text{Aut}(K_5)$. Let the vertices of X_5 form m_5 $\text{Aut}(X_5)$ -orbits. We take the middle point of the line segment joining vertices a_0 and c_1 as the origin $(0, 0)$ of K_5 (see Section 4). Let $\alpha_5 := a_6 - a_0$ and $\beta_5 := a_{32} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_5 := \langle x \mapsto x + \alpha_5, x \mapsto x + \beta_5 \rangle$ and

$$G_5 = \{ \alpha : x \mapsto \varepsilon x + m_5 \alpha_5 + n_5 \beta_5 : \varepsilon = \pm 1, m_5, n_5 \in \mathbb{Z} \} \cong H_5 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_5 \trianglelefteq G_5$ and the number of G_5/Γ_5 -orbits of vertices of X_5 is twelve. Therefore, G_5/Γ_5 acts on $X_5 = K_5/\Gamma_5$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$, $O_{11} = \langle c_3 \rangle$, $O_{12} = \langle c_4 \rangle$ are the G_5 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$, $O_{11} = \langle c_3 \rangle$, $O_{12} = \langle c_4 \rangle$ are the (G_5/Γ_5) -orbits. Since the vertex set of X_5 is $\sqcup_{j=1}^{12} \eta_5(O_j)$ and $G_5/\Gamma_5 \leq \text{Aut}(X_5)$, it follows that the number of $\text{Aut}(X_5)$ -orbits of vertices is ≤ 12 .

Let $X_6 = K_6/\Gamma_6$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_6 \leq \text{Aut}(K_6)$. Let the vertices of X_6 form m_6 $\text{Aut}(X_6)$ -orbits.

We take the middle point of the line segment joining vertices a_0 and a_1 as the origin $(0,0)$ of K_6 (see Section 4). Let $\alpha_6 := a_2 - a_0$ and $\beta_6 := a_{22} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_6 := \langle x \mapsto x + \alpha_6, x \mapsto x + \beta_6 \rangle$ and

$$G_6 = \{\alpha : x \mapsto \varepsilon x + m_6 \alpha_6 + n_6 \beta_6 : \varepsilon = \pm 1, m_6, n_6 \in \mathbb{Z}\} \cong H_6 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_6 \trianglelefteq G_6$ and the number of G_6/Γ_6 -orbits of vertices of X_6 is seven. Therefore, G_6/Γ_6 acts on $X_6 = K_6/\Gamma_6$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_9 \rangle$, $O_3 = \langle a_8 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_{10} \rangle$ are the G_6 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_9 \rangle$, $O_3 = \langle a_8 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_{10} \rangle$ are the (G_6/Γ_6) -orbits. Since the vertex set of X_6 is $\sqcup_{j=1}^7 \eta_6(O_j)$ and $G_6/\Gamma_6 \leq \text{Aut}(X_6)$, it follows that the number of $\text{Aut}(X_6)$ -orbits of vertices is ≤ 7 .

Let $X_7 = K_7/\Gamma_7$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_7 \leq \text{Aut}(K_7)$. Let the vertices of X_7 form m_7 $\text{Aut}(X_7)$ -orbits. We take the point a_0 as the origin $(0,0)$ of K_7 (see Section 4). Let $\alpha_7 := a_1 - a_0$, $\beta_7 := a_2 - a_0$ and $\gamma_7 := a_3 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_7 := \langle x \mapsto x + \alpha_7, x \mapsto x + \beta_7, x \mapsto x + \gamma_7 \rangle$ and

$$G_7 = \{\alpha : x \mapsto \varepsilon x + m_7 \alpha_7 + n_7 \beta_7 + r_7 \gamma_7 : \varepsilon = \pm 1, m_7, n_7, r_7 \in \mathbb{Z}\} \cong H_7 \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_7 \trianglelefteq G_7$ and the number of G_7/Γ_7 -orbits of vertices of X_7 is seven. Therefore, G_7/Γ_7 acts on $X_7 = K_7/\Gamma_7$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{19} \rangle$, $O_4 = \langle b_{17} \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_2 \rangle$ are the G_7 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{19} \rangle$, $O_4 = \langle b_{17} \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_2 \rangle$ are the (G_7/Γ_7) -orbits. Since the vertex set of X_7 is $\sqcup_{j=1}^7 \eta_7(O_j)$ and $G_7/\Gamma_7 \leq \text{Aut}(X_7)$, it follows that the number of $\text{Aut}(X_7)$ -orbits of vertices is ≤ 7 .

Let $X_8 = K_8/\Gamma_8$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_8 \leq \text{Aut}(K_8)$. Let the vertices of X_8 form m_8 $\text{Aut}(X_8)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_6 as the origin $(0,0)$ of K_8 (see Section 4). Let $\alpha_8 := a_{28} - a_0$ and $\beta_8 := a_{46} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_8 := \langle x \mapsto x + \alpha_8, x \mapsto x + \beta_8 \rangle$ and

$$G_8 = \{\alpha : x \mapsto \varepsilon x + m_8 \alpha_8 + n_8 \beta_8 : \varepsilon = \pm 1, m_8, n_8 \in \mathbb{Z}\} \cong H_8 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_8 \trianglelefteq G_8$ and the number of G_8/Γ_8 -orbits of vertices of X_8 is ten. Therefore, G_8/Γ_8 acts on $X_8 = K_8/\Gamma_8$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$ are the G_8 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$ are the (G_8/Γ_8) -orbits. Since the vertex set of X_8 is $\sqcup_{j=1}^{10} \eta_8(O_j)$ and $G_8/\Gamma_8 \leq \text{Aut}(X_8)$, it follows that the number of $\text{Aut}(X_8)$ -orbits of vertices is ≤ 10 .

Let $X_9 = K_9/\Gamma_9$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_9 \leq \text{Aut}(K_9)$. Let the vertices of X_9 form m_9 $\text{Aut}(X_9)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_9 (see Section 4). Let $\alpha_9 := a_8 - a_0$, $\beta_9 := a_{15} - a_0$ and $\gamma_9 := a_{18} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_9 := \langle x \mapsto x + \alpha_9, x \mapsto x + \beta_9, x \mapsto x + \gamma_9 \rangle$ and

$$G_9 = \{\alpha : x \mapsto \varepsilon x + m_9 \alpha_9 + n_9 \beta_9 + r_9 \gamma_9 : \varepsilon = \pm 1, m_9, n_9, r_9 \in \mathbb{Z}\} \cong H_9 \rtimes \mathbb{Z}_2$$

We take the middle point of the line segment joining vertices a_0 and a_1 as the origin $(0,0)$ of K_6 (see Section 4). Let $\alpha_6 := a_2 - a_0$ and $\beta_6 := a_{22} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_6 := \langle x \mapsto x + \alpha_6, x \mapsto x + \beta_6 \rangle$ and

$$G_6 = \{\alpha : x \mapsto \varepsilon x + m_6 \alpha_6 + n_6 \beta_6 : \varepsilon = \pm 1, m_6, n_6 \in \mathbb{Z}\} \cong H_6 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_6 \trianglelefteq G_6$ and the number of G_6/Γ_6 -orbits of vertices of X_6 is seven. Therefore, G_6/Γ_6 acts on $X_6 = K_6/\Gamma_6$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_9 \rangle$, $O_3 = \langle a_8 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_{10} \rangle$ are the G_6 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_9 \rangle$, $O_3 = \langle a_8 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_{10} \rangle$ are the (G_6/Γ_6) -orbits. Since the vertex set of X_6 is $\sqcup_{j=1}^7 \eta_6(O_j)$ and $G_6/\Gamma_6 \leq \text{Aut}(X_6)$, it follows that the number of $\text{Aut}(X_6)$ -orbits of vertices is ≤ 7 .

Let $X_7 = K_7/\Gamma_7$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_7 \leq \text{Aut}(K_7)$. Let the vertices of X_7 form m_7 $\text{Aut}(X_7)$ -orbits. We take the point a_0 as the origin $(0,0)$ of K_7 (see Section 4). Let $\alpha_7 := a_1 - a_0$, $\beta_7 := a_2 - a_0$ and $\gamma_7 := a_3 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_7 := \langle x \mapsto x + \alpha_7, x \mapsto x + \beta_7, x \mapsto x + \gamma_7 \rangle$ and

$$G_7 = \{\alpha : x \mapsto \varepsilon x + m_7 \alpha_7 + n_7 \beta_7 + r_7 \gamma_7 : \varepsilon = \pm 1, m_7, n_7, r_7 \in \mathbb{Z}\} \cong H_7 \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_7 \trianglelefteq G_7$ and the number of G_7/Γ_7 -orbits of vertices of X_7 is seven. Therefore, G_7/Γ_7 acts on $X_7 = K_7/\Gamma_7$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{19} \rangle$, $O_4 = \langle b_{17} \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_2 \rangle$ are the G_7 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{19} \rangle$, $O_4 = \langle b_{17} \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_2 \rangle$ are the (G_7/Γ_7) -orbits. Since the vertex set of X_7 is $\sqcup_{j=1}^7 \eta_7(O_j)$ and $G_7/\Gamma_7 \leq \text{Aut}(X_7)$, it follows that the number of $\text{Aut}(X_7)$ -orbits of vertices is ≤ 7 .

Let $X_8 = K_8/\Gamma_8$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_8 \leq \text{Aut}(K_8)$. Let the vertices of X_8 form m_8 $\text{Aut}(X_8)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_6 as the origin $(0,0)$ of K_8 (see Section 4). Let $\alpha_8 := a_{28} - a_0$ and $\beta_8 := a_{46} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_8 := \langle x \mapsto x + \alpha_8, x \mapsto x + \beta_8 \rangle$ and

$$G_8 = \{\alpha : x \mapsto \varepsilon x + m_8 \alpha_8 + n_8 \beta_8 : \varepsilon = \pm 1, m_8, n_8 \in \mathbb{Z}\} \cong H_8 \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_8 \trianglelefteq G_8$ and the number of G_8/Γ_8 -orbits of vertices of X_8 is ten. Therefore, G_8/Γ_8 acts on $X_8 = K_8/\Gamma_8$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$ are the G_8 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$ are the (G_8/Γ_8) -orbits. Since the vertex set of X_8 is $\sqcup_{j=1}^{10} \eta_8(O_j)$ and $G_8/\Gamma_8 \leq \text{Aut}(X_8)$, it follows that the number of $\text{Aut}(X_8)$ -orbits of vertices is ≤ 10 .

Let $X_9 = K_9/\Gamma_9$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_9 \leq \text{Aut}(K_9)$. Let the vertices of X_9 form m_9 $\text{Aut}(X_9)$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_9 (see Section 4). Let $\alpha_9 := a_8 - a_0$, $\beta_9 := a_{15} - a_0$ and $\gamma_9 := a_{18} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_9 := \langle x \mapsto x + \alpha_9, x \mapsto x + \beta_9, x \mapsto x + \gamma_9 \rangle$ and

$$G_9 = \{\alpha : x \mapsto \varepsilon x + m_9 \alpha_9 + n_9 \beta_9 + r_9 \gamma_9 : \varepsilon = \pm 1, m_9, n_9, r_9 \in \mathbb{Z}\} \cong H_9 \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_9 \trianglelefteq G_9$ and the number of G_9/Γ_9 -orbits of vertices of X_9 is ten. Therefore, G_9/Γ_9 acts on $X_9 = K_9/\Gamma_9$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_4 \rangle$, $O_8 = \langle c_5 \rangle$, $O_9 = \langle c_8 \rangle$, $O_{10} = \langle c_9 \rangle$ are the G_9 -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_4 \rangle$, $O_8 = \langle c_5 \rangle$, $O_9 = \langle c_8 \rangle$, $O_{10} = \langle c_9 \rangle$ are the (G_9/Γ_9) -orbits. Since the vertex set of X_9 is $\sqcup_{j=1}^{10} \eta_9(O_j)$ and $G_9/\Gamma_9 \leq \text{Aut}(X_9)$, it follows that the number of $\text{Aut}(X_9)$ -orbits of vertices is ≤ 10 .

Let $X_{10} = K_{10}/\Gamma_{10}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{10} \leq \text{Aut}(K_{10})$. Let the vertices of X_{10} form m_{10} $\text{Aut}(X_{10})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{13} as the origin $(0,0)$ of K_{10} (see Section 4). Let $\alpha_{10} := a_1 - a_0$ and $\beta_{10} := a_{22} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{10} := \langle x \mapsto x + \alpha_{10}, x \mapsto x + \beta_{10} \rangle$ and

$$G_{10} = \{ \alpha : x \mapsto \varepsilon x + m_{10}\alpha_{10} + n_{10}\beta_{10} : \varepsilon = \pm 1, m_{10}, n_{10} \in \mathbb{Z} \} \cong H_{10} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{10} \trianglelefteq G_{10}$ and the number of G_{10}/Γ_{10} -orbits of vertices of X_{10} is three. Therefore, G_{10}/Γ_{10} acts on $X_{10} = K_{10}/\Gamma_{10}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_0 \rangle$ are the G_{10} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_0 \rangle$ are the (G_{10}/Γ_{10}) -orbits. Since the vertex set of X_{10} is $\sqcup_{j=1}^3 \eta_{10}(O_j)$ and $G_{10}/\Gamma_{10} \leq \text{Aut}(X_{10})$, it follows that the number of $\text{Aut}(X_{10})$ -orbits of vertices is 3.

Let $X_{11} = K_{11}/\Gamma_{11}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{11} \leq \text{Aut}(K_{11})$. Let the vertices of X_{11} form m_{11} $\text{Aut}(X_{11})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{13} as the origin $(0,0)$ of K_{11} (see Section 4). Let $\alpha_{11} := a_1 - a_0$ and $\beta_{11} := a_{22} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{11} := \langle x \mapsto x + \alpha_{11}, x \mapsto x + \beta_{11} \rangle$ and

$$G_{11} = \{ \alpha : x \mapsto \varepsilon x + m_{11}\alpha_{11} + n_{11}\beta_{11} : \varepsilon = \pm 1, m_{11}, n_{11} \in \mathbb{Z} \} \cong H_{11} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{11} \trianglelefteq G_{11}$ and the number of G_{11}/Γ_{11} -orbits of vertices of X_{11} is three. Therefore, G_{11}/Γ_{11} acts on $X_{11} = K_{11}/\Gamma_{11}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_0 \rangle$ are the G_{11} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_0 \rangle$ are the (G_{11}/Γ_{11}) -orbits. Since the vertex set of X_{11} is $\sqcup_{j=1}^3 \eta_{11}(O_j)$ and $G_{11}/\Gamma_{11} \leq \text{Aut}(X_{11})$, it follows that the number of $\text{Aut}(X_{11})$ -orbits of vertices is 3.

Let $X_{12} = K_{12}/\Gamma_{12}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{12} \leq \text{Aut}(K_{12})$. Let the vertices of X_{12} form m_{12} $\text{Aut}(X_{12})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{11} as the origin $(0,0)$ of K_{12} (see Section 4). Let $\alpha_{12} := a_1 - a_0$ and $\beta_{12} := a_{19} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{12} := \langle x \mapsto x + \alpha_{12}, x \mapsto x + \beta_{12} \rangle$ and

$$G_{12} = \{ \alpha : x \mapsto \varepsilon x + m_{12}\alpha_{12} + n_{12}\beta_{12} : \varepsilon = \pm 1, m_{12}, n_{12} \in \mathbb{Z} \} \cong H_{12} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{12} \trianglelefteq G_{12}$ and the number of G_{12}/Γ_{12} -orbits of vertices of X_{12} is three. Therefore, G_{12}/Γ_{12} acts on $X_{12} = K_{12}/\Gamma_{12}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_2 \rangle$, $O_3 = \langle c_0 \rangle$ are the G_{12} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_2 \rangle$, $O_3 = \langle c_0 \rangle$ are the (G_{12}/Γ_{12}) -orbits. Since the vertex set of X_{12} is $\sqcup_{j=1}^3 \eta_{12}(O_j)$ and $G_{12}/\Gamma_{12} \leq \text{Aut}(X_{12})$, it follows that the number of $\text{Aut}(X_{12})$ -orbits of vertices is 3.

Let $X_{13} = K_{13}/\Gamma_{13}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{13} \leq \text{Aut}(K_{13})$. Let the vertices of X_{13} form m_{13} $\text{Aut}(X_{13})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{14} as the origin

$(0, 0)$ of K_{13} (see Section 4). Let $\alpha_{13} := a_1 - a_0$ and $\beta_{13} := a_{24} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{13} := \langle x \mapsto x + \alpha_{13}, x \mapsto x + \beta_{13} \rangle$ and

$$G_{13} = \{\alpha : x \mapsto \varepsilon x + m_{13}\alpha_{13} + n_{13}\beta_{13} : \varepsilon = \pm 1, m_{13}, n_{13} \in \mathbb{Z}\} \cong H_{13} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{13} \trianglelefteq G_{13}$ and the number of G_{13}/Γ_{13} -orbits of vertices of X_{13} is three. Therefore, G_{13}/Γ_{13} acts on $X_{13} = K_{13}/\Gamma_{13}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_2 \rangle$ are the G_{13} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_3 \rangle$, $O_3 = \langle c_2 \rangle$ are the (G_{13}/Γ_{13}) -orbits. Since the vertex set of X_{13} is $\sqcup_{j=1}^3 \eta_{13}(O_j)$ and $G_{13}/\Gamma_{13} \leq \text{Aut}(X_{13})$, it follows that the number of $\text{Aut}(X_{13})$ -orbits of vertices is 3.

Let $X_{15} = K_{15}/\Gamma_{15}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{15} \leq \text{Aut}(K_{15})$. Let the vertices of X_{15} form m_{15} $\text{Aut}(X_{15})$ -orbits. We take the middle point of the line segment joining vertices c_0 and c_3 as the origin $(0, 0)$ of K_{15} (see Section 4). Let $\alpha_{15} := a_1 - a_0$, $\beta_{15} := a_2 - a_0$ and $\gamma_{15} := a_3 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{15} := \langle x \mapsto x + \alpha_{15}, x \mapsto x + \beta_{15}, x \mapsto x + \gamma_{15} \rangle$ and

$$G_{15} = \{\alpha : x \mapsto \varepsilon x + m_{15}\alpha_{15} + n_{15}\beta_{15} + r_{15}\gamma_{15} : \varepsilon = \pm 1, m_{15}, n_{15}, r_{15} \in \mathbb{Z}\} \cong H_{15} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{15} \trianglelefteq G_{15}$ and the number of G_{15}/Γ_{15} -orbits of vertices of X_7 is ten. Therefore, G_{15}/Γ_{15} acts on $X_{15} = K_{15}/\Gamma_{15}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_{16} \rangle$, $O_5 = \langle b_{15} \rangle$, $O_6 = \langle b_{14} \rangle$, $O_7 = \langle b_{13} \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$ are the G_{15} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_{16} \rangle$, $O_5 = \langle b_{15} \rangle$, $O_6 = \langle b_{14} \rangle$, $O_7 = \langle b_{13} \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$ are the (G_{15}/Γ_{15}) -orbits. Since the vertex set of X_{15} is $\sqcup_{j=1}^{10} \eta_{15}(O_j)$ and $G_{15}/\Gamma_{15} \leq \text{Aut}(X_{15})$, it follows that the number of $\text{Aut}(X_{15})$ -orbits of vertices is ≤ 10 .

Let $X_{16} = K_{16}/\Gamma_{16}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{16} \leq \text{Aut}(K_{16})$. Let the vertices of X_{16} form m_{16} $\text{Aut}(X_{16})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_3 as the origin $(0, 0)$ of K_{16} (see Section 4). Let $\alpha_{16} := a_1 - a_0$, $\beta_{16} := a_2 - a_0$ and $\gamma_{16} := a_3 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{16} := \langle x \mapsto x + \alpha_{16}, x \mapsto x + \beta_{16}, x \mapsto x + \gamma_{16} \rangle$ and

$$G_{16} = \{\alpha : x \mapsto \varepsilon x + m_{16}\alpha_{16} + n_{16}\beta_{16} + r_{16}\gamma_{16} : \varepsilon = \pm 1, m_{16}, n_{16}, r_{16} \in \mathbb{Z}\} \cong H_{16} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{16} \trianglelefteq G_{16}$ and the number of G_{16}/Γ_{16} -orbits of vertices of X_{16} is eleven. Therefore, G_{16}/Γ_{16} acts on $X_{16} = K_{16}/\Gamma_{16}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_2 \rangle$, $O_5 = \langle b_3 \rangle$, $O_6 = \langle c_0 \rangle$, $O_7 = \langle c_1 \rangle$, $O_8 = \langle c_2 \rangle$, $O_9 = \langle c_3 \rangle$, $O_{10} = \langle c_4 \rangle$, $O_{11} = \langle c_5 \rangle$ are the G_{16} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_2 \rangle$, $O_5 = \langle b_3 \rangle$, $O_6 = \langle c_0 \rangle$, $O_7 = \langle c_1 \rangle$, $O_8 = \langle c_2 \rangle$, $O_9 = \langle c_3 \rangle$, $O_{10} = \langle c_4 \rangle$, $O_{11} = \langle c_5 \rangle$ are the (G_{16}/Γ_{16}) -orbits. Since the vertex set of X_{16} is $\sqcup_{j=1}^{11} \eta_{16}(O_j)$ and $G_{16}/\Gamma_{16} \leq \text{Aut}(X_{16})$, it follows that the number of $\text{Aut}(X_{16})$ -orbits of vertices is ≤ 11 .

Let $X_{17} = K_{17}/\Gamma_{17}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{17} \leq \text{Aut}(K_{17})$. Let the vertices of X_{17} form m_{17} $\text{Aut}(X_{17})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0, 0)$ of K_{17} (see Section 4). Let $\alpha_{17} := a_6 - a_0$, $\beta_{17} := a_{15} - a_0$ and $\gamma_{17} := a_{21} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{17} := \langle x \mapsto x + \alpha_{17}, x \mapsto x + \beta_{17}, x \mapsto x + \gamma_{17} \rangle$ and

$$G_{17} = \{\alpha : x \mapsto \varepsilon x + m_{17}\alpha_{17} + n_{17}\beta_{17} + r_{17}\gamma_{17} : \varepsilon = \pm 1, m_{17}, n_{17}, r_{17} \in \mathbb{Z}\} \cong H_{17} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{17} \trianglelefteq G_{17}$ and the number of G_{17}/Γ_{17} -orbits of vertices of X_{17} is ten. Therefore, G_{17}/Γ_{17} acts on $X_{17} = K_{17}/\Gamma_{17}$. Since $O_1 = \langle a_0 \rangle$,

$O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_{15} \rangle$, $O_8 = \langle c_{16} \rangle$, $O_9 = \langle c_{17} \rangle$, $O_{10} = \langle c_{18} \rangle$ are the G_{17} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$, $O_6 = \langle c_1 \rangle$, $O_7 = \langle c_{15} \rangle$, $O_8 = \langle c_{16} \rangle$, $O_9 = \langle c_{17} \rangle$, $O_{10} = \langle c_{18} \rangle$ are the (G_{17}/Γ_{17}) -orbits. Since the vertex set of X_{17} is $\sqcup_{j=1}^{10} \eta_{17}(O_j)$ and $G_{17}/\Gamma_{17} \leq \text{Aut}(X_{17})$, it follows that the number of $\text{Aut}(X_{17})$ -orbits of vertices is ≤ 10 .

Let $X_{18} = K_{18}/\Gamma_{18}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{18} \leq \text{Aut}(K_{18})$. Let the vertices of X_{18} form m_{18} $\text{Aut}(X_{18})$ -orbits. We take the middle point of the line segment joining vertices a_0 and c_2 as the origin $(0,0)$ of K_{18} (see Section 4). Let $\alpha_{18} := a_{11} - a_0$, $\beta_{18} := a_{18} - a_0$ and $\gamma_{18} := a_{31} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{18} := \langle x \mapsto x + \alpha_{18}, x \mapsto x + \beta_{18}, x \mapsto x + \gamma_{18} \rangle$ and

$$G_{18} = \{ \alpha : x \mapsto \varepsilon x + m_{18}\alpha_{18} + n_{18}\beta_{18} + r_{18}\gamma_{18} : \varepsilon = \pm 1, m_{18}, n_{18}, r_{18} \in \mathbb{Z} \} \cong H_{18} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{18} \trianglelefteq G_{18}$ and the number of G_{18}/Γ_{18} -orbits of vertices of X_{18} is thirteen. Therefore, G_{18}/Γ_{18} acts on $X_{18} = K_{18}/\Gamma_{18}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$, $O_{11} = \langle c_3 \rangle$, $O_{12} = \langle c_4 \rangle$, $O_{13} = \langle c_5 \rangle$ are the G_{18} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle c_0 \rangle$, $O_9 = \langle c_1 \rangle$, $O_{10} = \langle c_2 \rangle$, $O_{11} = \langle c_3 \rangle$, $O_{12} = \langle c_4 \rangle$, $O_{13} = \langle c_5 \rangle$ are the (G_{18}/Γ_{18}) -orbits. Since the vertex set of X_{18} is $\sqcup_{j=1}^{13} \eta_{18}(O_j)$ and $G_{18}/\Gamma_{18} \leq \text{Aut}(X_{18})$, it follows that the number of $\text{Aut}(X_{18})$ -orbits of vertices is ≤ 13 .

Let $X_{19} = K_{19}/\Gamma_{19}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{19} \leq \text{Aut}(K_{19})$. Let the vertices of X_{19} form m_{19} $\text{Aut}(X_{19})$ -orbits. We take the point b_0 as the origin $(0,0)$ of K_{19} (see Section 4). Let $\alpha_{19} := b_1 - b_0$, $\beta_{19} := b_2 - b_0$ and $\gamma_{19} := b_3 - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{19} := \langle x \mapsto x + \alpha_{19}, x \mapsto x + \beta_{19}, x \mapsto x + \gamma_{19} \rangle$ and

$$G_{19} = \{ \alpha : x \mapsto \varepsilon x + m_{19}\alpha_{19} + n_{19}\beta_{19} + r_{19}\gamma_{19} : \varepsilon = \pm 1, m_{19}, n_{19}, r_{19} \in \mathbb{Z} \} \cong H_{19} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{19} \trianglelefteq G_{19}$ and the number of G_{19}/Γ_{19} -orbits of vertices of X_{19} is five. Therefore, G_{19}/Γ_{19} acts on $X_{19} = K_{19}/\Gamma_{19}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_5 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$ are the G_{19} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_5 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle c_0 \rangle$ are the (G_{19}/Γ_{19}) -orbits. Since the vertex set of X_{19} is $\sqcup_{j=1}^5 \eta_{19}(O_j)$ and $G_{19}/\Gamma_{19} \leq \text{Aut}(X_{19})$, it follows that the number of $\text{Aut}(X_{19})$ -orbits of vertices is ≤ 5 .

Let $X_{20} = K_{20}/\Gamma_{20}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{20} \leq \text{Aut}(K_{20})$. Let the vertices of X_{20} form m_{20} $\text{Aut}(X_{20})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{20} (see Section 4). Let $\alpha_{20} := a_6 - a_0$, $\beta_{20} := a_{12} - a_0$ and $\gamma_{20} := a_{18} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{20} := \langle x \mapsto x + \alpha_{20}, x \mapsto x + \beta_{20}, x \mapsto x + \gamma_{20} \rangle$ and

$$G_{20} = \{ \alpha : x \mapsto \varepsilon x + m_{20}\alpha_{20} + n_{20}\beta_{20} + r_{20}\gamma_{20} : \varepsilon = \pm 1, m_{20}, n_{20}, r_{20} \in \mathbb{Z} \} \cong H_{20} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{20} \trianglelefteq G_{20}$ and the number of G_{20}/Γ_{20} -orbits of vertices of X_{20} is nine. Therefore, G_{20}/Γ_{20} acts on $X_{20} = K_{20}/\Gamma_{20}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_4 \rangle$, $O_5 = \langle b_6 \rangle$, $O_6 = \langle b_8 \rangle$, $O_7 = \langle c_3 \rangle$, $O_8 = \langle c_5 \rangle$, $O_9 = \langle c_6 \rangle$ are the G_{20} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_4 \rangle$, $O_5 = \langle b_6 \rangle$, $O_6 = \langle b_8 \rangle$, $O_7 = \langle c_3 \rangle$, $O_8 = \langle c_5 \rangle$, $O_9 = \langle c_6 \rangle$ are the (G_{20}/Γ_{20}) -orbits. Since the vertex set of X_{20} is $\sqcup_{j=1}^9 \eta_{20}(O_j)$ and $G_{20}/\Gamma_{20} \leq \text{Aut}(X_{20})$, it follows that the number of $\text{Aut}(X_{20})$ -orbits of vertices is ≤ 9 .

Let $X_{21} = K_{21}/\Gamma_{21}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{21} \leq \text{Aut}(K_{21})$. Let the vertices of X_{21} form m_{21} $\text{Aut}(X_{21})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_1 as the origin $(0,0)$ of K_{21} (see Section 4). Let $\alpha_{21} := a_2 - a_0$ and $\beta_{21} := a_8 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{21} := \langle x \mapsto x + \alpha_{21}, x \mapsto x + \beta_{21} \rangle$ and

$$G_{21} = \{ \alpha : x \mapsto \varepsilon x + m_{21}\alpha_{21} + n_{21}\beta_{21} : \varepsilon = \pm 1, m_{21}, n_{21} \in \mathbb{Z} \} \cong H_{21} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{21} \trianglelefteq G_{21}$ and the number of G_{21}/Γ_{21} -orbits of vertices of X_{21} is four. Therefore, G_{21}/Γ_{21} acts on $X_{21} = K_{21}/\Gamma_{21}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle c_0 \rangle$, $O_3 = \langle c_1 \rangle$, $O_4 = \langle b_0 \rangle$ are the G_{21} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle c_0 \rangle$, $O_3 = \langle c_1 \rangle$, $O_4 = \langle b_0 \rangle$ are the (G_{21}/Γ_{21}) -orbits. Since the vertex set of X_{21} is $\sqcup_{j=1}^4 \eta_{21}(O_j)$ and $G_{21}/\Gamma_{21} \leq \text{Aut}(X_{21})$, it follows that the number of $\text{Aut}(X_{21})$ -orbits of vertices is ≤ 4 .

Let $X_{22} = K_{22}/\Gamma_{22}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{22} \leq \text{Aut}(K_{22})$. Let the vertices of X_{22} form m_{22} $\text{Aut}(X_{22})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_1 as the origin $(0,0)$ of K_{22} (see Section 4). Let $\alpha_{22} := a_1 - a_0$ and $\beta_{22} := a_2 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{22} := \langle x \mapsto x + \alpha_{22}, x \mapsto x + \beta_{22} \rangle$ and

$$G_{22} = \{ \alpha : x \mapsto \varepsilon x + m_{22}\alpha_{22} + n_{22}\beta_{22} : \varepsilon = \pm 1, m_{22}, n_{22} \in \mathbb{Z} \} \cong H_{22} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{22} \trianglelefteq G_{22}$ and the number of G_{22}/Γ_{22} -orbits of vertices of X_{22} is six. Therefore, G_{22}/Γ_{22} acts on $X_{22} = K_{22}/\Gamma_{22}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle c_0 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle c_8 \rangle$, $O_6 = \langle a_{10} \rangle$ are the G_{22} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle c_0 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle c_8 \rangle$, $O_6 = \langle a_{10} \rangle$ are the (G_{22}/Γ_{22}) -orbits. Since the vertex set of X_{22} is $\sqcup_{j=1}^6 \eta_{22}(O_j)$ and $G_{22}/\Gamma_{22} \leq \text{Aut}(X_{22})$, it follows that the number of $\text{Aut}(X_{22})$ -orbits of vertices is ≤ 6 .

Let $X_{23} = K_{23}/\Gamma_{23}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{23} \leq \text{Aut}(K_{23})$. Let the vertices of X_{23} form m_{23} $\text{Aut}(X_{23})$ -orbits. We take the point b_0 as the origin $(0,0)$ of K_{23} (see Section 4). Let $\alpha_{23} := b_1 - b_0$ and $\beta_{23} := b_2 - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{23} := \langle x \mapsto x + \alpha_{23}, x \mapsto x + \beta_{23} \rangle$ and

$$G_{23} = \{ \alpha : x \mapsto \varepsilon x + m_{23}\alpha_{23} + n_{23}\beta_{23} : \varepsilon = \pm 1, m_{23}, n_{23} \in \mathbb{Z} \} \cong H_{23} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{23} \trianglelefteq G_{23}$ and the number of G_{23}/Γ_{23} -orbits of vertices of X_{23} is five. Therefore, G_{23}/Γ_{23} acts on $X_{23} = K_{23}/\Gamma_{23}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle c_0 \rangle$, $O_5 = \langle c_1 \rangle$ are the G_{23} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle c_0 \rangle$, $O_5 = \langle c_1 \rangle$ are the (G_{23}/Γ_{23}) -orbits. Since the vertex set of X_{23} is $\sqcup_{j=1}^5 \eta_{23}(O_j)$ and $G_{23}/\Gamma_{23} \leq \text{Aut}(X_{23})$, it follows that the number of $\text{Aut}(X_{23})$ -orbits of vertices is ≤ 5 .

Let $X_{24} = K_{24}/\Gamma_{24}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{24} \leq \text{Aut}(K_{24})$. Let the vertices of X_{24} form m_{24} $\text{Aut}(X_{24})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_6 as the origin $(0,0)$ of K_{24} (see Section 4). Let $\alpha_{24} := a_{12} - a_0$, $\beta_{24} := a_{30} - a_0$ and $\gamma_{24} := a_{43} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{24} := \langle x \mapsto x + \alpha_{24}, x \mapsto x + \beta_{24}, x \mapsto x + \gamma_{24} \rangle$ and

$$G_{24} = \{ \alpha : x \mapsto \varepsilon x + m_{24}\alpha_{24} + n_{24}\beta_{24} + r_{24}\gamma_{24} : \varepsilon = \pm 1, m_{24}, n_{24}, r_{24} \in \mathbb{Z} \} \cong H_{24} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{24} \trianglelefteq G_{24}$ and the number of G_{24}/Γ_{24} -orbits of vertices of X_{24} is fifteen. Therefore, G_{24}/Γ_{24} acts on $X_{24} = K_{24}/\Gamma_{24}$. Since $O_1 = \langle a_0 \rangle$,

$O_2 = \langle a_1 \rangle, O_3 = \langle a_2 \rangle, O_4 = \langle a_3 \rangle, O_5 = \langle a_4 \rangle, O_6 = \langle a_5 \rangle, O_7 = \langle b_0 \rangle, O_8 = \langle b_2 \rangle, O_9 = \langle b_4 \rangle, O_{10} = \langle c_0 \rangle, O_{11} = \langle c_1 \rangle, O_{12} = \langle c_2 \rangle, O_{13} = \langle c_3 \rangle, O_{14} = \langle c_4 \rangle, O_{15} = \langle c_5 \rangle$ are the G_{24} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle a_2 \rangle, O_4 = \langle a_3 \rangle, O_5 = \langle a_4 \rangle, O_6 = \langle a_5 \rangle, O_7 = \langle b_0 \rangle, O_8 = \langle b_2 \rangle, O_9 = \langle b_4 \rangle, O_{10} = \langle c_0 \rangle, O_{11} = \langle c_1 \rangle, O_{12} = \langle c_2 \rangle, O_{13} = \langle c_3 \rangle, O_{14} = \langle c_4 \rangle, O_{15} = \langle c_5 \rangle$ are the (G_{24}/Γ_{24}) -orbits. Since the vertex set of X_{24} is $\sqcup_{j=1}^{15} \eta_{24}(O_j)$ and $G_{24}/\Gamma_{24} \leq \text{Aut}(X_{24})$, it follows that the number of $\text{Aut}(X_{24})$ -orbits of vertices is ≤ 15 .

Let $X_{25} = K_{25}/\Gamma_{25}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{25} \leq \text{Aut}(K_{25})$. Let the vertices of X_{25} form m_{25} $\text{Aut}(X_{25})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_1 as the origin $(0,0)$ of K_{25} (see Section 4). Let $\alpha_{25} := b_2 - b_0$ and $\beta_{25} := b_5 - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{25} := \langle x \mapsto x + \alpha_{25}, x \mapsto x + \beta_{25} \rangle$ and

$$G_{25} = \{ \alpha : x \mapsto \varepsilon x + m_{25}\alpha_{25} + n_{25}\beta_{25} : \varepsilon = \pm 1, m_{25}, n_{25} \in \mathbb{Z} \} \cong H_{25} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{25} \trianglelefteq G_{25}$ and the number of G_{25}/Γ_{25} -orbits of vertices of X_{25} is five. Therefore, G_{25}/Γ_{25} acts on $X_{25} = K_{25}/\Gamma_{25}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle b_0 \rangle, O_4 = \langle c_0 \rangle, O_5 = \langle c_2 \rangle$ are the G_{25} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle b_0 \rangle, O_4 = \langle c_0 \rangle, O_5 = \langle c_2 \rangle$ are the (G_{25}/Γ_{25}) -orbits. Since the vertex set of X_{25} is $\sqcup_{j=1}^5 \eta_{25}(O_j)$ and $G_{25}/\Gamma_{25} \leq \text{Aut}(X_{25})$, it follows that the number of $\text{Aut}(X_{25})$ -orbits of vertices is ≤ 5 .

Let $X_{26} = K_{26}/\Gamma_{26}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{26} \leq \text{Aut}(K_{26})$. Let the vertices of X_{26} form m_{26} $\text{Aut}(X_{26})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_1 as the origin $(0,0)$ of K_{26} (see Section 4). Let $\alpha_{26} := b_3 - b_0$ and $\beta_{26} := b_{12} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{26} := \langle x \mapsto x + \alpha_{26}, x \mapsto x + \beta_{26} \rangle$ and

$$G_{26} = \{ \alpha : x \mapsto \varepsilon x + m_{26}\alpha_{26} + n_{26}\beta_{26} : \varepsilon = \pm 1, m_{26}, n_{26} \in \mathbb{Z} \} \cong H_{26} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{26} \trianglelefteq G_{26}$ and the number of G_{26}/Γ_{26} -orbits of vertices of X_{26} is four. Therefore, G_{26}/Γ_{26} acts on $X_{26} = K_{26}/\Gamma_{26}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle b_0 \rangle, O_3 = \langle c_{10} \rangle, O_4 = \langle c_{11} \rangle$ are the G_{26} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle b_0 \rangle, O_3 = \langle c_{10} \rangle, O_4 = \langle c_{11} \rangle$ are the (G_{26}/Γ_{26}) -orbits. Since the vertex set of X_{26} is $\sqcup_{j=1}^4 \eta_{26}(O_j)$ and $G_{26}/\Gamma_{26} \leq \text{Aut}(X_{26})$, it follows that the number of $\text{Aut}(X_{26})$ -orbits of vertices is ≤ 4 .

Let $X_{27} = K_{27}/\Gamma_{27}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{27} \leq \text{Aut}(K_{27})$. Let the vertices of X_{27} form m_{27} $\text{Aut}(X_{27})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_2 as the origin $(0,0)$ of K_{27} (see Section 4). Let $\alpha_{27} := a_5 - a_0$ and $\beta_{27} := a_{15} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{27} := \langle x \mapsto x + \alpha_{27}, x \mapsto x + \beta_{27} \rangle$ and

$$G_{27} = \{ \alpha : x \mapsto \varepsilon x + m_{27}\alpha_{27} + n_{27}\beta_{27} : \varepsilon = \pm 1, m_{27}, n_{27} \in \mathbb{Z} \} \cong H_{27} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{27} \trianglelefteq G_{27}$ and the number of G_{27}/Γ_{27} -orbits of vertices of X_{27} is five. Therefore, G_{27}/Γ_{27} acts on $X_{27} = K_{27}/\Gamma_{27}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle b_0 \rangle, O_4 = \langle c_0 \rangle, O_5 = \langle c_1 \rangle$ are the G_{27} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle b_0 \rangle, O_4 = \langle c_0 \rangle, O_5 = \langle c_1 \rangle$ are the (G_{27}/Γ_{27}) -orbits. Since the vertex set of X_{27} is $\sqcup_{j=1}^5 \eta_{27}(O_j)$ and $G_{27}/\Gamma_{27} \leq \text{Aut}(X_{27})$, it follows that the number of $\text{Aut}(X_{27})$ -orbits of vertices is ≤ 5 .

Let $X_{29} = K_{29}/\Gamma_{29}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{29} \leq \text{Aut}(K_{29})$. Let the vertices of X_{29} form m_{29} $\text{Aut}(X_{29})$ -orbits. We take the middle point of the line segment joining vertices a_0 and c_1 as the origin

$(0,0)$ of K_{29} (see Section 4). Let $\alpha_{29} := a_2 - a_0$ and $\beta_{29} := a_{13} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{29} := \langle x \mapsto x + \alpha_{29}, x \mapsto x + \beta_{29} \rangle$ and

$$G_{29} = \{ \alpha : x \mapsto \varepsilon x + m_{29}\alpha_{29} + n_{29}\beta_{29} : \varepsilon = \pm 1, m_{29}, n_{29} \in \mathbb{Z} \} \cong H_{29} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{29} \trianglelefteq G_{29}$ and the number of G_{29}/Γ_{29} -orbits of vertices of X_{29} is nine. Therefore, G_{29}/Γ_{29} acts on $X_{29} = K_{29}/\Gamma_{29}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_7 \rangle$, $O_5 = \langle b_8 \rangle$, $O_6 = \langle c_0 \rangle$, $O_7 = \langle c_1 \rangle$, $O_8 = \langle c_{36} \rangle$, $O_9 = \langle c_{37} \rangle$ are the G_{29} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle b_7 \rangle$, $O_5 = \langle b_8 \rangle$, $O_6 = \langle c_0 \rangle$, $O_7 = \langle c_1 \rangle$, $O_8 = \langle c_{36} \rangle$, $O_9 = \langle c_{37} \rangle$ are the (G_{29}/Γ_{29}) -orbits. Since the vertex set of X_{29} is $\sqcup_{j=1}^9 \eta_{29}(O_j)$ and $G_{29}/\Gamma_{29} \leq \text{Aut}(X_{29})$, it follows that the number of $\text{Aut}(X_{29})$ -orbits of vertices is ≤ 9 .

Let $X_{30} = K_{30}/\Gamma_{30}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{30} \leq \text{Aut}(K_{30})$. Let the vertices of X_{30} form m_{30} $\text{Aut}(X_{30})$ -orbits. We take the middle point of the line segment joining vertices a_0 and c_1 as the origin $(0,0)$ of K_{30} (see Section 4). Let $\alpha_{30} := a_2 - a_0$ and $\beta_{30} := a_{38} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{30} := \langle x \mapsto x + \alpha_{30}, x \mapsto x + \beta_{30} \rangle$ and

$$G_{30} = \{ \alpha : x \mapsto \varepsilon x + m_{30}\alpha_{30} + n_{30}\beta_{30} : \varepsilon = \pm 1, m_{30}, n_{30} \in \mathbb{Z} \} \cong H_{30} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{30} \trianglelefteq G_{30}$ and the number of G_{30}/Γ_{30} -orbits of vertices of X_{30} is eight. Therefore, G_{30}/Γ_{30} acts on $X_{30} = K_{30}/\Gamma_{30}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_{27} \rangle$, $O_4 = \langle a_{28} \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_9 \rangle$, $O_7 = \langle c_0 \rangle$, $O_8 = \langle c_1 \rangle$ are the G_{30} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_{27} \rangle$, $O_4 = \langle a_{28} \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_9 \rangle$, $O_7 = \langle c_0 \rangle$, $O_8 = \langle c_1 \rangle$ are the (G_{30}/Γ_{30}) -orbits. Since the vertex set of X_{30} is $\sqcup_{j=1}^8 \eta_{30}(O_j)$ and $G_{30}/\Gamma_{30} \leq \text{Aut}(X_{30})$, it follows that the number of $\text{Aut}(X_{30})$ -orbits of vertices is ≤ 8 .

Let $X_{31} = K_{31}/\Gamma_{31}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{31} \leq \text{Aut}(K_{31})$. Let the vertices of X_{31} form m_{31} $\text{Aut}(X_{31})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_1 as the origin $(0,0)$ of K_{31} (see Section 4). Let $\alpha_{31} := a_1 - a_0$ and $\beta_{31} := a_2 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{31} := \langle x \mapsto x + \alpha_{31}, x \mapsto x + \beta_{31} \rangle$ and

$$G_{31} = \{ \alpha : x \mapsto \varepsilon x + m_{31}\alpha_{31} + n_{31}\beta_{31} : \varepsilon = \pm 1, m_{31}, n_{31} \in \mathbb{Z} \} \cong H_{31} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{31} \trianglelefteq G_{31}$ and the number of G_{31}/Γ_{31} -orbits of vertices of X_{31} is six. Therefore, G_{31}/Γ_{31} acts on $X_{31} = K_{31}/\Gamma_{31}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle c_0 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle c_{14} \rangle$, $O_6 = \langle a_1 \rangle$ are the G_{31} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle c_0 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle c_{14} \rangle$, $O_6 = \langle a_1 \rangle$ are the (G_{31}/Γ_{31}) -orbits. Since the vertex set of X_{31} is $\sqcup_{j=1}^6 \eta_{31}(O_j)$ and $G_{31}/\Gamma_{31} \leq \text{Aut}(X_{31})$, it follows that the number of $\text{Aut}(X_{31})$ -orbits of vertices is ≤ 6 .

Let $X_{32} = K_{32}/\Gamma_{32}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{32} \leq \text{Aut}(K_{32})$. Let the vertices of X_{32} form m_{32} $\text{Aut}(X_{32})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{32} (see Section 4). Let $\alpha_{32} := a_6 - a_0$, $\beta_{32} := a_{20} - a_0$ and $\gamma_{32} := a_{27} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{32} := \langle x \mapsto x + \alpha_{32}, x \mapsto x + \beta_{32}, x \mapsto x + \gamma_{32} \rangle$ and

$$G_{32} = \{ \alpha : x \mapsto \varepsilon x + m_{32}\alpha_{32} + n_{32}\beta_{32} + r_{32}\gamma_{32} : \varepsilon = \pm 1, m_{32}, n_{32}, r_{32} \in \mathbb{Z} \} \cong H_{32} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{32} \trianglelefteq G_{32}$ and the number of G_{32}/Γ_{32} -orbits of vertices of X_{32} is twelve. Therefore, G_{32}/Γ_{32} acts on $X_{32} = K_{32}/\Gamma_{32}$. Since $O_1 = \langle a_0 \rangle$,

$O_2 = \langle a_1 \rangle, O_3 = \langle a_2 \rangle, O_4 = \langle c_0 \rangle, O_5 = \langle b_0 \rangle, O_6 = \langle b_1 \rangle, O_7 = \langle c_1 \rangle, O_8 = \langle b_2 \rangle, O_9 = \langle b_3 \rangle, O_{10} = \langle c_2 \rangle, O_{11} = \langle b_4 \rangle, O_{12} = \langle b_5 \rangle$ are the G_{32} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle a_2 \rangle, O_4 = \langle c_0 \rangle, O_5 = \langle b_0 \rangle, O_6 = \langle b_1 \rangle, O_7 = \langle c_1 \rangle, O_8 = \langle b_2 \rangle, O_9 = \langle b_3 \rangle, O_{10} = \langle c_2 \rangle, O_{11} = \langle b_4 \rangle, O_{12} = \langle b_5 \rangle$ are the (G_{32}/Γ_{32}) -orbits. Since the vertex set of X_{32} is $\sqcup_{j=1}^{12} \eta_{32}(O_j)$ and $G_{32}/\Gamma_{32} \leq \text{Aut}(X_{32})$, it follows that the number of $\text{Aut}(X_{32})$ -orbits of vertices is ≤ 12 .

Let $X_{33} = K_{33}/\Gamma_{33}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{33} \leq \text{Aut}(K_{33})$. Let the vertices of X_{33} form m_{33} $\text{Aut}(X_{33})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{33} (see Section 4). Let $\alpha_{33} := a_6 - a_0, \beta_{33} := a_{12} - a_0$ and $\gamma_{33} := a_{18} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{33} := \langle x \mapsto x + \alpha_{33}, x \mapsto x + \beta_{33}, x \mapsto x + \gamma_{33} \rangle$ and

$$G_{33} = \{ \alpha : x \mapsto \varepsilon x + m_{33}\alpha_{33} + n_{33}\beta_{33} + r_{33}\gamma_{33} : \varepsilon = \pm 1, m_{33}, n_{33}, r_{33} \in \mathbb{Z} \} \cong H_{32} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{33} \trianglelefteq G_{33}$ and the number of G_{33}/Γ_{33} -orbits of vertices of X_{33} is twelve. Therefore, G_{32}/Γ_{33} acts on $X_{33} = K_{33}/\Gamma_{33}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle a_2 \rangle, O_4 = \langle b_0 \rangle, O_5 = \langle c_0 \rangle, O_6 = \langle c_1 \rangle, O_7 = \langle b_2 \rangle, O_8 = \langle c_2 \rangle, O_9 = \langle c_3 \rangle, O_{10} = \langle b_4 \rangle, O_{11} = \langle c_4 \rangle, O_{12} = \langle c_5 \rangle$ are the G_{33} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle a_2 \rangle, O_4 = \langle b_0 \rangle, O_5 = \langle c_0 \rangle, O_6 = \langle c_1 \rangle, O_7 = \langle b_2 \rangle, O_8 = \langle c_2 \rangle, O_9 = \langle c_3 \rangle, O_{10} = \langle b_4 \rangle, O_{11} = \langle c_4 \rangle, O_{12} = \langle c_5 \rangle$ are the (G_{33}/Γ_{33}) -orbits. Since the vertex set of X_{33} is $\sqcup_{j=1}^{12} \eta_{33}(O_j)$ and $G_{33}/\Gamma_{33} \leq \text{Aut}(X_{33})$, it follows that the number of $\text{Aut}(X_{33})$ -orbits of vertices is ≤ 12 .

Let $X_{35} = K_{35}/\Gamma_{35}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{35} \leq \text{Aut}(K_{35})$. Let the vertices of X_{35} form m_{35} $\text{Aut}(X_{35})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_2 as the origin $(0,0)$ of K_{35} (see Section 4). Let $\alpha_{35} := a_{12} - a_0$ and $\beta_{35} := a_{43} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{35} := \langle x \mapsto x + \alpha_{35}, x \mapsto x + \beta_{35} \rangle$ and

$$G_{35} = \{ \alpha : x \mapsto \varepsilon x + m_{35}\alpha_{35} + n_{35}\beta_{35} : \varepsilon = \pm 1, m_{35}, n_{35} \in \mathbb{Z} \} \cong H_{35} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{35} \trianglelefteq G_{35}$ and the number of G_{35}/Γ_{35} -orbits of vertices of X_{35} is four. Therefore, G_{35}/Γ_{35} acts on $X_{35} = K_{35}/\Gamma_{35}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle c_1 \rangle, O_4 = \langle b_0 \rangle$ are the G_{35} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle c_1 \rangle, O_4 = \langle b_0 \rangle$ are the (G_{35}/Γ_{35}) -orbits. Since the vertex set of X_{35} is $\sqcup_{j=1}^4 \eta_{35}(O_j)$ and $G_{35}/\Gamma_{35} \leq \text{Aut}(X_{35})$, it follows that the number of $\text{Aut}(X_{35})$ -orbits of vertices is ≤ 4 .

Let $X_{36} = K_{36}/\Gamma_{36}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{36} \leq \text{Aut}(K_{36})$. Let the vertices of X_{36} form m_{36} $\text{Aut}(X_{36})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_2 as the origin $(0,0)$ of K_{36} (see Section 4). Let $\alpha_{36} := a_{11} - a_0$ and $\beta_{36} := a_{35} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{36} := \langle x \mapsto x + \alpha_{36}, x \mapsto x + \beta_{36} \rangle$ and

$$G_{36} = \{ \alpha : x \mapsto \varepsilon x + m_{36}\alpha_{36} + n_{36}\beta_{36} : \varepsilon = \pm 1, m_{36}, n_{36} \in \mathbb{Z} \} \cong H_{36} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{36} \trianglelefteq G_{36}$ and the number of G_{36}/Γ_{36} -orbits of vertices of X_{36} is five. Therefore, G_{36}/Γ_{36} acts on $X_{36} = K_{36}/\Gamma_{36}$. Since $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle c_1 \rangle, O_4 = \langle b_0 \rangle, O_5 = \langle b_1 \rangle$ are the G_{36} -orbits, it follows that $O_1 = \langle a_0 \rangle, O_2 = \langle a_1 \rangle, O_3 = \langle c_1 \rangle, O_4 = \langle b_0 \rangle, O_5 = \langle b_1 \rangle$ are the (G_{36}/Γ_{36}) -orbits. Since the vertex set of X_{36} is $\sqcup_{j=1}^5 \eta_{36}(O_j)$ and $G_{36}/\Gamma_{36} \leq \text{Aut}(X_{36})$, it follows that the number of $\text{Aut}(X_{36})$ -orbits of vertices is ≤ 5 .

Let $X_{37} = K_{37}/\Gamma_{37}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{37} \leq \text{Aut}(K_{37})$. Let the vertices of X_{37} form m_{37} $\text{Aut}(X_{37})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_2 as the origin

$(0, 0)$ of K_{37} (see Section 4). Let $\alpha_{37} := a_{12} - a_0$ and $\beta_{37} := a_{42} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{37} := \langle x \mapsto x + \alpha_{37}, x \mapsto x + \beta_{37} \rangle$ and

$$G_{37} = \{ \alpha : x \mapsto \varepsilon x + m_{37}\alpha_{37} + n_{37}\beta_{37} : \varepsilon = \pm 1, m_{37}, n_{37} \in \mathbb{Z} \} \cong H_{37} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{37} \trianglelefteq G_{37}$ and the number of G_{37}/Γ_{37} -orbits of vertices of X_{37} is five. Therefore, G_{37}/Γ_{37} acts on $X_{37} = K_{37}/\Gamma_{37}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle c_1 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_1 \rangle$ are the G_{37} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle c_1 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle b_1 \rangle$ are the (G_{37}/Γ_{37}) -orbits. Since the vertex set of X_{37} is $\sqcup_{j=1}^5 \eta_{37}(O_j)$ and $G_{37}/\Gamma_{37} \leq \text{Aut}(X_{37})$, it follows that the number of $\text{Aut}(X_{37})$ -orbits of vertices is ≤ 5 .

Let $X_{38} = K_{38}/\Gamma_{38}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{38} \leq \text{Aut}(K_{38})$. Let the vertices of X_{38} form m_{38} $\text{Aut}(X_{38})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_2 as the origin $(0, 0)$ of K_{38} (see Section 4). Let $\alpha_{38} := a_{10} - a_0$ and $\beta_{38} := a_{35} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{38} := \langle x \mapsto x + \alpha_{38}, x \mapsto x + \beta_{38} \rangle$ and

$$G_{38} = \{ \alpha : x \mapsto \varepsilon x + m_{38}\alpha_{38} + n_{38}\beta_{38} : \varepsilon = \pm 1, m_{38}, n_{38} \in \mathbb{Z} \} \cong H_{38} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{38} \trianglelefteq G_{38}$ and the number of G_{38}/Γ_{38} -orbits of vertices of X_{38} is five. Therefore, G_{38}/Γ_{38} acts on $X_{38} = K_{38}/\Gamma_{38}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{23} \rangle$, $O_4 = \langle a_1 \rangle$, $O_5 = \langle c_1 \rangle$ are the G_{38} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{23} \rangle$, $O_4 = \langle a_1 \rangle$, $O_5 = \langle c_1 \rangle$ are the (G_{38}/Γ_{38}) -orbits. Since the vertex set of X_{38} is $\sqcup_{j=1}^5 \eta_{38}(O_j)$ and $G_{38}/\Gamma_{38} \leq \text{Aut}(X_{38})$, it follows that the number of $\text{Aut}(X_{38})$ -orbits of vertices is ≤ 5 .

Let $X_{39} = K_{39}/\Gamma_{39}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{39} \leq \text{Aut}(K_{39})$. Let the vertices of X_{39} form m_{39} $\text{Aut}(X_{39})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_6 as the origin $(0, 0)$ of K_{39} (see Section 4). Let $\alpha_{39} := a_{12} - a_0$, $\beta_{39} := a_{36} - a_0$ and $\gamma_{39} := a_{52} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{39} := \langle x \mapsto x + \alpha_{39}, x \mapsto x + \beta_{39}, x \mapsto x + \gamma_{39} \rangle$ and

$$G_{39} = \{ \alpha : x \mapsto \varepsilon x + m_{39}\alpha_{39} + n_{39}\beta_{39} + r_{39}\gamma_{39} : \varepsilon = \pm 1, m_{39}, n_{39}, r_{39} \in \mathbb{Z} \} \cong H_{39} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{39} \trianglelefteq G_{39}$ and the number of G_{39}/Γ_{39} -orbits of vertices of X_{39} is fifteen. Therefore, G_{39}/Γ_{39} acts on $X_{39} = K_{39}/\Gamma_{39}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle c_0 \rangle$, $O_8 = \langle b_0 \rangle$, $O_9 = \langle b_1 \rangle$, $O_{10} = \langle c_4 \rangle$, $O_{11} = \langle c_5 \rangle$, $O_{12} = \langle b_5 \rangle$, $O_{13} = \langle c_8 \rangle$, $O_{14} = \langle c_{11} \rangle$, $O_{15} = \langle c_{12} \rangle$ are the G_{39} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_3 \rangle$, $O_5 = \langle a_4 \rangle$, $O_6 = \langle a_5 \rangle$, $O_7 = \langle c_0 \rangle$, $O_8 = \langle b_0 \rangle$, $O_9 = \langle b_1 \rangle$, $O_{10} = \langle c_4 \rangle$, $O_{11} = \langle c_5 \rangle$, $O_{12} = \langle b_5 \rangle$, $O_{13} = \langle c_8 \rangle$, $O_{14} = \langle c_{11} \rangle$, $O_{15} = \langle c_{12} \rangle$ are the (G_{39}/Γ_{39}) -orbits. Since the vertex set of X_{39} is $\sqcup_{j=1}^{15} \eta_{39}(O_j)$ and $G_{39}/\Gamma_{39} \leq \text{Aut}(X_{39})$, it follows that the number of $\text{Aut}(X_{39})$ -orbits of vertices is ≤ 15 .

Let $X_{40} = K_{40}/\Gamma_{40}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{40} \leq \text{Aut}(K_{40})$. Let the vertices of X_{40} form m_{40} $\text{Aut}(X_{40})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0, 0)$ of K_{40} (see Section 4). Let $\alpha_{40} := a_7 - a_0$ and $\beta_{40} := a_{15} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{40} := \langle x \mapsto x + \alpha_{40}, x \mapsto x + \beta_{40} \rangle$ and

$$G_{40} = \{ \alpha : x \mapsto \varepsilon x + m_{40}\alpha_{40} + n_{40}\beta_{40} : \varepsilon = \pm 1, m_{40}, n_{40} \in \mathbb{Z} \} \cong H_{40} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{40} \trianglelefteq G_{40}$ and the number of G_{40}/Γ_{40} -orbits of vertices of X_{40} is seven. Therefore, G_{40}/Γ_{40} acts on $X_{40} = K_{40}/\Gamma_{40}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_{13} \rangle$, $O_5 = \langle b_{12} \rangle$, $O_6 = \langle b_{10} \rangle$, $O_7 = \langle b_{11} \rangle$ are the G_{40} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_{13} \rangle$, $O_5 = \langle b_{12} \rangle$, $O_6 = \langle b_{10} \rangle$, $O_7 = \langle b_{11} \rangle$ are the (G_{40}/Γ_{40}) -orbits. Since the vertex set of X_{40} is $\sqcup_{j=1}^7 \eta_{40}(O_j)$ and $G_{40}/\Gamma_{40} \leq \text{Aut}(X_{40})$, it follows that the number of $\text{Aut}(X_{40})$ -orbits of vertices is ≤ 7 .

Let $X_{41} = K_{41}/\Gamma_{41}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{41} \leq \text{Aut}(K_{41})$. Let the vertices of X_{41} form m_{41} $\text{Aut}(X_{41})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_6 as the origin $(0,0)$ of K_{41} (see Section 4). Let $\alpha_{41} := a_{11} - a_0$, $\beta_{41} := a_{15} - a_0$ and $\gamma_{41} := a_{21} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{41} := \langle x \mapsto x + \alpha_{41}, x \mapsto x + \beta_{41}, x \mapsto x + \gamma_{41} \rangle$ and

$$G_{41} = \{ \alpha : x \mapsto \varepsilon x + m_{41}\alpha_{41} + n_{41}\beta_{41} + r_{41}\gamma_{41} : \varepsilon = \pm 1, m_{41}, n_{41}, r_{41} \in \mathbb{Z} \} \cong H_{41} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{41} \trianglelefteq G_{41}$ and the number of G_{41}/Γ_{41} -orbits of vertices of X_{41} is nine. Therefore, G_{41}/Γ_{41} acts on $X_{41} = K_{41}/\Gamma_{41}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_{18} \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_1 \rangle$, $O_7 = \langle b_2 \rangle$, $O_8 = \langle b_{19} \rangle$, $O_9 = \langle b_{24} \rangle$ are the G_{41} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_{18} \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_1 \rangle$, $O_7 = \langle b_2 \rangle$, $O_8 = \langle b_{19} \rangle$, $O_9 = \langle b_{24} \rangle$ are the (G_{41}/Γ_{41}) -orbits. Since the vertex set of X_{41} is $\sqcup_{j=1}^9 \eta_{41}(O_j)$ and $G_{41}/\Gamma_{41} \leq \text{Aut}(X_{41})$, it follows that the number of $\text{Aut}(X_{41})$ -orbits of vertices is ≤ 9 .

Let $X_{42} = K_{42}/\Gamma_{42}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{42} \leq \text{Aut}(K_{42})$. Let the vertices of X_{42} form m_{42} $\text{Aut}(X_{42})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{42} (see Section 4). Let $\alpha_{42} := a_{11} - a_0$, $\beta_{42} := a_{15} - a_0$ and $\gamma_{42} := a_{21} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{42} := \langle x \mapsto x + \alpha_{42}, x \mapsto x + \beta_{42}, x \mapsto x + \gamma_{42} \rangle$ and

$$G_{42} = \{ \alpha : x \mapsto \varepsilon x + m_{42}\alpha_{42} + n_{42}\beta_{42} + r_{42}\gamma_{42} : \varepsilon = \pm 1, m_{42}, n_{42}, r_{42} \in \mathbb{Z} \} \cong H_{42} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{42} \trianglelefteq G_{42}$ and the number of G_{42}/Γ_{42} -orbits of vertices of X_{42} is nine. Therefore, G_{42}/Γ_{42} acts on $X_{42} = K_{42}/\Gamma_{42}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_{24} \rangle$, $O_5 = \langle b_{25} \rangle$, $O_6 = \langle b_{14} \rangle$, $O_7 = \langle b_5 \rangle$, $O_8 = \langle b_{47} \rangle$, $O_9 = \langle b_{46} \rangle$ are the G_{42} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_{24} \rangle$, $O_5 = \langle b_{25} \rangle$, $O_6 = \langle b_{14} \rangle$, $O_7 = \langle b_5 \rangle$, $O_8 = \langle b_{47} \rangle$, $O_9 = \langle b_{46} \rangle$ are the (G_{42}/Γ_{42}) -orbits. Since the vertex set of X_{42} is $\sqcup_{j=1}^9 \eta_{42}(O_j)$ and $G_{42}/\Gamma_{42} \leq \text{Aut}(X_{42})$, it follows that the number of $\text{Aut}(X_{42})$ -orbits of vertices is ≤ 9 .

Let $X_{43} = K_{43}/\Gamma_{43}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{43} \leq \text{Aut}(K_{43})$. Let the vertices of X_{43} form m_{43} $\text{Aut}(X_{43})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{43} (see Section 4). Let $\alpha_{43} := a_7 - a_0$ and $\beta_{43} := a_{13} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{43} := \langle x \mapsto x + \alpha_{43}, x \mapsto x + \beta_{43} \rangle$ and

$$G_{43} = \{ \alpha : x \mapsto \varepsilon x + m_{43}\alpha_{43} + n_{43}\beta_{43} : \varepsilon = \pm 1, m_{43}, n_{43} \in \mathbb{Z} \} \cong H_{43} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{43} \trianglelefteq G_{43}$ and the number of G_{43}/Γ_{43} -orbits of vertices of X_{43} is seven. Therefore, G_{43}/Γ_{43} acts on $X_{43} = K_{43}/\Gamma_{43}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_{24} \rangle$, $O_5 = \langle a_{25} \rangle$, $O_6 = \langle a_{26} \rangle$, $O_7 = \langle b_0 \rangle$ are the G_{43} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_{24} \rangle$, $O_5 = \langle a_{25} \rangle$, $O_6 = \langle a_{26} \rangle$, $O_7 = \langle b_0 \rangle$ are the (G_{43}/Γ_{43}) -orbits. Since the vertex set of X_{43} is $\sqcup_{j=1}^7 \eta_{43}(O_j)$ and $G_{43}/\Gamma_{43} \leq \text{Aut}(X_{43})$, it follows that the number of $\text{Aut}(X_{43})$ -orbits of vertices is ≤ 7 .

Let $X_{44} = K_{44}/\Gamma_{44}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{44} \leq \text{Aut}(K_{44})$. Let the vertices of X_{44} form m_{44} $\text{Aut}(X_{44})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_{18} as the origin $(0,0)$ of K_{44} (see Section 4). Let $\alpha_{44} := b_1 - b_0$ and $\beta_{44} := b_{30} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{44} := \langle x \mapsto x + \alpha_{44}, x \mapsto x + \beta_{44} \rangle$ and

$$G_{44} = \{ \alpha : x \mapsto \varepsilon x + m_{44}\alpha_{44} + n_{44}\beta_{44} : \varepsilon = \pm 1, m_{44}, n_{44} \in \mathbb{Z} \} \cong H_{44} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{44} \trianglelefteq G_{44}$ and the number of G_{44}/Γ_{44} -orbits of vertices of X_{44} is three. Therefore, G_{44}/Γ_{44} acts on $X_{44} = K_{44}/\Gamma_{44}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_{18} \rangle$, $O_3 = \langle b_0 \rangle$ are the G_{44} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_{18} \rangle$, $O_3 = \langle b_0 \rangle$ are the (G_{44}/Γ_{44}) -orbits. Since the vertex set of X_{44} is $\sqcup_{j=1}^3 \eta_{44}(O_j)$ and $G_{44}/\Gamma_{44} \leq \text{Aut}(X_{44})$, it follows that the number of $\text{Aut}(X_{44})$ -orbits of vertices is 3.

Let $X_{45} = K_{45}/\Gamma_{45}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{45} \leq \text{Aut}(K_{45})$. Let the vertices of X_{45} form m_{45} $\text{Aut}(X_{45})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_{11} as the origin $(0,0)$ of K_{45} (see Section 4). Let $\alpha_{45} := b_1 - b_0$ and $\beta_{45} := b_{28} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{45} := \langle x \mapsto x + \alpha_{45}, x \mapsto x + \beta_{45} \rangle$ and

$$G_{45} = \{ \alpha : x \mapsto \varepsilon x + m_{45}\alpha_{45} + n_{45}\beta_{45} : \varepsilon = \pm 1, m_{45}, n_{45} \in \mathbb{Z} \} \cong H_{45} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{45} \trianglelefteq G_{45}$ and the number of G_{45}/Γ_{45} -orbits of vertices of X_{45} is four. Therefore, G_{45}/Γ_{45} acts on $X_{45} = K_{45}/\Gamma_{45}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_{11} \rangle$, $O_3 = \langle a_{21} \rangle$, $O_4 = \langle b_0 \rangle$ are the G_{45} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_{11} \rangle$, $O_3 = \langle a_{21} \rangle$, $O_4 = \langle b_0 \rangle$ are the (G_{45}/Γ_{45}) -orbits. Since the vertex set of X_{45} is $\sqcup_{j=1}^4 \eta_{45}(O_j)$ and $G_{45}/\Gamma_{45} \leq \text{Aut}(X_{45})$, it follows that the number of $\text{Aut}(X_{45})$ -orbits of vertices is ≤ 4 .

Let $X_{46} = K_{46}/\Gamma_{46}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{46} \leq \text{Aut}(K_{46})$. Let the vertices of X_{46} form m_{46} $\text{Aut}(X_{46})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_{14} as the origin $(0,0)$ of K_{46} (see Section 4). Let $\alpha_{46} := b_1 - b_0$ and $\beta_{46} := b_{85} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{46} := \langle x \mapsto x + \alpha_{46}, x \mapsto x + \beta_{46} \rangle$ and

$$G_{46} = \{ \alpha : x \mapsto \varepsilon x + m_{46}\alpha_{46} + n_{46}\beta_{46} : \varepsilon = \pm 1, m_{46}, n_{46} \in \mathbb{Z} \} \cong H_{46} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{46} \trianglelefteq G_{46}$ and the number of G_{46}/Γ_{46} -orbits of vertices of X_{46} is four. Therefore, G_{46}/Γ_{46} acts on $X_{46} = K_{46}/\Gamma_{46}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{60} \rangle$, $O_4 = \langle b_{75} \rangle$ are the G_{46} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle b_{60} \rangle$, $O_4 = \langle b_{75} \rangle$ are the (G_{46}/Γ_{46}) -orbits. Since the vertex set of X_{46} is $\sqcup_{j=1}^4 \eta_{46}(O_j)$ and $G_{46}/\Gamma_{46} \leq \text{Aut}(X_{46})$, it follows that the number of $\text{Aut}(X_{46})$ -orbits of vertices is ≤ 4 .

Let $X_{47} = K_{47}/\Gamma_{47}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{47} \leq \text{Aut}(K_{47})$. Let the vertices of X_{47} form m_{47} $\text{Aut}(X_{47})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_{16} as the origin $(0,0)$ of K_{47} (see Section 4). Let $\alpha_{47} := b_1 - b_0$ and $\beta_{47} := b_{51} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{47} := \langle x \mapsto x + \alpha_{47}, x \mapsto x + \beta_{47} \rangle$ and

$$G_{47} = \{ \alpha : x \mapsto \varepsilon x + m_{47}\alpha_{47} + n_{47}\beta_{47} : \varepsilon = \pm 1, m_{47}, n_{47} \in \mathbb{Z} \} \cong H_{47} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{47} \trianglelefteq G_{47}$ and the number of G_{47}/Γ_{47} -orbits of vertices of X_{47} is five. Therefore, G_{47}/Γ_{47} acts on $X_{47} = K_{47}/\Gamma_{47}$. Since $O_1 = \langle b_0 \rangle$,

$O_2 = \langle b_{28} \rangle$, $O_3 = \langle b_{39} \rangle$, $O_4 = \langle a_0 \rangle$, $O_5 = \langle a_{16} \rangle$ are the G_{47} -orbits, it follows that $O_1 = \langle b_0 \rangle$, $O_2 = \langle b_{28} \rangle$, $O_3 = \langle b_{39} \rangle$, $O_4 = \langle a_0 \rangle$, $O_5 = \langle a_{16} \rangle$ are the (G_{47}/Γ_{47}) -orbits. Since the vertex set of X_{47} is $\sqcup_{j=1}^5 \eta_{47}(O_j)$ and $G_{47}/\Gamma_{47} \leq \text{Aut}(X_{47})$, it follows that the number of $\text{Aut}(X_{47})$ -orbits of vertices is ≤ 5 .

Let $X_{48} = K_{48}/\Gamma_{48}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{48} \leq \text{Aut}(K_{48})$. Let the vertices of X_{48} form m_{48} $\text{Aut}(X_{48})$ -orbits. We take the point b_0 as the origin $(0, 0)$ of K_{48} (see Section 4). Let $\alpha_{48} := b_1 - b_0$, $\beta_{48} := b_2 - b_0$ and $\gamma_{48} := b_3 - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{48} := \langle x \mapsto x + \alpha_{48}, x \mapsto x + \beta_{48}, x \mapsto x + \gamma_{48} \rangle$ and

$$G_{48} = \{ \alpha : x \mapsto \varepsilon x + m_{48}\alpha_{48} + n_{48}\beta_{48} + r_{48}\gamma_{48} : \varepsilon = \pm 1, m_{48}, n_{48}, r_{48} \in \mathbb{Z} \} \cong H_{48} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{48} \trianglelefteq G_{48}$ and the number of G_{48}/Γ_{48} -orbits of vertices of X_{48} is seven. Therefore, G_{48}/Γ_{48} acts on $X_{48} = K_{48}/\Gamma_{48}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_6 \rangle$, $O_5 = \langle a_8 \rangle$, $O_6 = \langle a_{10} \rangle$, $O_7 = \langle b_0 \rangle$ are the G_{48} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_6 \rangle$, $O_5 = \langle a_8 \rangle$, $O_6 = \langle a_{10} \rangle$, $O_7 = \langle b_0 \rangle$ are the (G_{48}/Γ_{48}) -orbits. Since the vertex set of X_{48} is $\sqcup_{j=1}^7 \eta_{48}(O_j)$ and $G_{48}/\Gamma_{48} \leq \text{Aut}(X_{48})$, it follows that the number of $\text{Aut}(X_{48})$ -orbits of vertices is ≤ 7 .

Let $X_{49} = K_{49}/\Gamma_{49}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{49} \leq \text{Aut}(K_{49})$. Let the vertices of X_{49} form m_{49} $\text{Aut}(X_{49})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_1 as the origin $(0, 0)$ of K_{49} (see Section 4). Let $\alpha_{49} := b_1 - b_0$ and $\beta_{49} := b_{11} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{49} := \langle x \mapsto x + \alpha_{49}, x \mapsto x + \beta_{49} \rangle$ and

$$G_{49} = \{ \alpha : x \mapsto \varepsilon x + m_{49}\alpha_{49} + n_{49}\beta_{49} : \varepsilon = \pm 1, m_{49}, n_{49} \in \mathbb{Z} \} \cong H_{49} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{49} \trianglelefteq G_{49}$ and the number of G_{49}/Γ_{49} -orbits of vertices of X_{49} is three. Therefore, G_{49}/Γ_{49} acts on $X_{49} = K_{49}/\Gamma_{49}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_0 \rangle$ are the G_{49} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_0 \rangle$ are the (G_{49}/Γ_{49}) -orbits. Since the vertex set of X_{49} is $\sqcup_{j=1}^3 \eta_{49}(O_j)$ and $G_{49}/\Gamma_{49} \leq \text{Aut}(X_{49})$, it follows that the number of $\text{Aut}(X_{49})$ -orbits of vertices is 3.

Let $X_{50} = K_{50}/\Gamma_{50}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{50} \leq \text{Aut}(K_{50})$. Let the vertices of X_{50} form m_{50} $\text{Aut}(X_{50})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_3 as the origin $(0, 0)$ of K_{50} (see Section 4). Let $\alpha_{50} := b_1 - b_0$ and $\beta_{50} := b_{10} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{50} := \langle x \mapsto x + \alpha_{50}, x \mapsto x + \beta_{50} \rangle$ and

$$G_{50} = \{ \alpha : x \mapsto \varepsilon x + m_{50}\alpha_{50} + n_{50}\beta_{50} : \varepsilon = \pm 1, m_{50}, n_{50} \in \mathbb{Z} \} \cong H_{50} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{50} \trianglelefteq G_{50}$ and the number of G_{50}/Γ_{50} -orbits of vertices of X_{50} is three. Therefore, G_{50}/Γ_{50} acts on $X_{50} = K_{50}/\Gamma_{50}$. Since $O_1 = \langle a_5 \rangle$, $O_2 = \langle a_6 \rangle$, $O_3 = \langle b_0 \rangle$ are the G_{50} -orbits, it follows that $O_1 = \langle a_5 \rangle$, $O_2 = \langle a_6 \rangle$, $O_3 = \langle b_0 \rangle$ are the (G_{50}/Γ_{50}) -orbits. Since the vertex set of X_{50} is $\sqcup_{j=1}^3 \eta_{50}(O_j)$ and $G_{50}/\Gamma_{50} \leq \text{Aut}(X_{50})$, it follows that the number of $\text{Aut}(X_{50})$ -orbits of vertices is 3.

Let $X_{51} = K_{51}/\Gamma_{51}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{51} \leq \text{Aut}(K_{51})$. Let the vertices of X_{51} form m_{51} $\text{Aut}(X_{51})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_1 as the origin

$(0, 0)$ of K_{51} (see Section 4). Let $\alpha_{51} := a_{14} - a_0$ and $\beta_{51} := a_7 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{51} := \langle x \mapsto x + \alpha_{51}, x \mapsto x + \beta_{51} \rangle$ and

$$G_{51} = \{ \alpha : x \mapsto \varepsilon x + m_{51}\alpha_{51} + n_{51}\beta_{51} : \varepsilon = \pm 1, m_{51}, n_{51} \in \mathbb{Z} \} \cong H_{51} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{51} \trianglelefteq G_{51}$ and the number of G_{51}/Γ_{51} -orbits of vertices of X_{51} is three. Therefore, G_{51}/Γ_{51} acts on $X_{51} = K_{51}/\Gamma_{51}$. Since $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_2 \rangle$, $O_3 = \langle b_{21} \rangle$, $O_4 = \langle b_{20} \rangle$, $O_5 = \langle a_3 \rangle$, $O_6 = \langle a_0 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle b_1 \rangle$, $O_9 = \langle b_2 \rangle$, $O_{10} = \langle a_8 \rangle$, $O_{11} = \langle a_{10} \rangle$, $O_{12} = \langle b_3 \rangle$, $O_{13} = \langle b_4 \rangle$, $O_{14} = \langle a_{11} \rangle$, $O_{15} = \langle a_9 \rangle$ are the G_{50} -orbits, it follows that $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_2 \rangle$, $O_3 = \langle b_{21} \rangle$, $O_4 = \langle b_{20} \rangle$, $O_5 = \langle a_3 \rangle$, $O_6 = \langle a_0 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle b_1 \rangle$, $O_9 = \langle b_2 \rangle$, $O_{10} = \langle a_8 \rangle$, $O_{11} = \langle a_{10} \rangle$, $O_{12} = \langle b_3 \rangle$, $O_{13} = \langle b_4 \rangle$, $O_{14} = \langle a_{11} \rangle$, $O_{15} = \langle a_9 \rangle$ are the (G_{51}/Γ_{51}) -orbits. Since the vertex set of X_{51} is $\sqcup_{j=1}^{15} \eta_{51}(O_j)$ and $G_{51}/\Gamma_{51} \leq \text{Aut}(X_{51})$, it follows that the number of $\text{Aut}(X_{51})$ -orbits of vertices is ≤ 15 .

Let $X_{52} = K_{52}/\Gamma_{52}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{52} \leq \text{Aut}(K_{52})$. Let the vertices of X_{52} form m_{52} $\text{Aut}(X_{52})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_0 as the origin $(0, 0)$ of K_{52} (see Section 4). Let $\alpha_{52} := a_3 - a_0$ and $\beta_{52} := a_5 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{52} := \langle x \mapsto x + \alpha_{52}, x \mapsto x + \beta_{52} \rangle$ and

$$G_{52} = \{ \alpha : x \mapsto \varepsilon x + m_{52}\alpha_{52} + n_{52}\beta_{52} : \varepsilon = \pm 1, m_{52}, n_{52} \in \mathbb{Z} \} \cong H_{52} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{52} \trianglelefteq G_{52}$ and the number of G_{52}/Γ_{52} -orbits of vertices of X_{52} is six. Therefore, G_{52}/Γ_{52} acts on $X_{52} = K_{52}/\Gamma_{52}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_1 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_3 \rangle$, $O_6 = \langle b_2 \rangle$ are the G_{52} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_1 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_3 \rangle$, $O_6 = \langle b_2 \rangle$ are the (G_{52}/Γ_{52}) -orbits. Since the vertex set of X_{52} is $\sqcup_{j=1}^6 \eta_{52}(O_j)$ and $G_{52}/\Gamma_{52} \leq \text{Aut}(X_{52})$, it follows that the number of $\text{Aut}(X_{52})$ -orbits of vertices is ≤ 6 .

Let $X_{53} = K_{53}/\Gamma_{53}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{53} \leq \text{Aut}(K_{53})$. Let the vertices of X_{53} form m_{53} $\text{Aut}(X_{53})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_1 as the origin $(0, 0)$ of K_{53} (see Section 4). Let $\alpha_{53} := a_1 - a_0$ and $\beta_{53} := a_{42} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{53} := \langle x \mapsto x + \alpha_{53}, x \mapsto x + \beta_{53} \rangle$ and

$$G_{53} = \{ \alpha : x \mapsto \varepsilon x + m_{53}\alpha_{53} + n_{53}\beta_{53} : \varepsilon = \pm 1, m_{53}, n_{53} \in \mathbb{Z} \} \cong H_{53} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{53} \trianglelefteq G_{53}$ and the number of G_{53}/Γ_{53} -orbits of vertices of X_{53} is five. Therefore, G_{53}/Γ_{53} acts on $X_{53} = K_{53}/\Gamma_{53}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{23} \rangle$, $O_5 = \langle a_{32} \rangle$ are the G_{53} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{23} \rangle$, $O_5 = \langle a_{32} \rangle$ are the (G_{53}/Γ_{53}) -orbits. Since the vertex set of X_{53} is $\sqcup_{j=1}^5 \eta_{53}(O_j)$ and $G_{53}/\Gamma_{53} \leq \text{Aut}(X_{53})$, it follows that the number of $\text{Aut}(X_{53})$ -orbits of vertices is ≤ 5 .

Let $X_{54} = K_{54}/\Gamma_{54}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{54} \leq \text{Aut}(K_{54})$. Let the vertices of X_{54} form m_{54} $\text{Aut}(X_{54})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{14} as the origin $(0, 0)$ of K_{54} (see Section 4). Let $\alpha_{54} := a_1 - a_0$ and $\beta_{54} := a_{81} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{54} := \langle x \mapsto x + \alpha_{54}, x \mapsto x + \beta_{54} \rangle$ and

$$G_{54} = \{ \alpha : x \mapsto \varepsilon x + m_{54}\alpha_{54} + n_{54}\beta_{54} : \varepsilon = \pm 1, m_{54}, n_{54} \in \mathbb{Z} \} \cong H_{54} \rtimes \mathbb{Z}_2.$$

$(0, 0)$ of K_{51} (see Section 4). Let $\alpha_{51} := a_{14} - a_0$ and $\beta_{51} := a_7 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{51} := \langle x \mapsto x + \alpha_{51}, x \mapsto x + \beta_{51} \rangle$ and

$$G_{51} = \{ \alpha : x \mapsto \varepsilon x + m_{51}\alpha_{51} + n_{51}\beta_{51} : \varepsilon = \pm 1, m_{51}, n_{51} \in \mathbb{Z} \} \cong H_{51} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{51} \trianglelefteq G_{51}$ and the number of G_{51}/Γ_{51} -orbits of vertices of X_{51} is three. Therefore, G_{51}/Γ_{51} acts on $X_{51} = K_{51}/\Gamma_{51}$. Since $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_2 \rangle$, $O_3 = \langle b_{21} \rangle$, $O_4 = \langle b_{20} \rangle$, $O_5 = \langle a_3 \rangle$, $O_6 = \langle a_0 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle b_1 \rangle$, $O_9 = \langle b_2 \rangle$, $O_{10} = \langle a_8 \rangle$, $O_{11} = \langle a_{10} \rangle$, $O_{12} = \langle b_3 \rangle$, $O_{13} = \langle b_4 \rangle$, $O_{14} = \langle a_{11} \rangle$, $O_{15} = \langle a_9 \rangle$ are the G_{50} -orbits, it follows that $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_2 \rangle$, $O_3 = \langle b_{21} \rangle$, $O_4 = \langle b_{20} \rangle$, $O_5 = \langle a_3 \rangle$, $O_6 = \langle a_0 \rangle$, $O_7 = \langle b_0 \rangle$, $O_8 = \langle b_1 \rangle$, $O_9 = \langle b_2 \rangle$, $O_{10} = \langle a_8 \rangle$, $O_{11} = \langle a_{10} \rangle$, $O_{12} = \langle b_3 \rangle$, $O_{13} = \langle b_4 \rangle$, $O_{14} = \langle a_{11} \rangle$, $O_{15} = \langle a_9 \rangle$ are the (G_{51}/Γ_{51}) -orbits. Since the vertex set of X_{51} is $\sqcup_{j=1}^{15} \eta_{51}(O_j)$ and $G_{51}/\Gamma_{51} \leq \text{Aut}(X_{51})$, it follows that the number of $\text{Aut}(X_{51})$ -orbits of vertices is ≤ 15 .

Let $X_{52} = K_{52}/\Gamma_{52}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{52} \leq \text{Aut}(K_{52})$. Let the vertices of X_{52} form m_{52} $\text{Aut}(X_{52})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_0 as the origin $(0, 0)$ of K_{52} (see Section 4). Let $\alpha_{52} := a_3 - a_0$ and $\beta_{52} := a_5 - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{52} := \langle x \mapsto x + \alpha_{52}, x \mapsto x + \beta_{52} \rangle$ and

$$G_{52} = \{ \alpha : x \mapsto \varepsilon x + m_{52}\alpha_{52} + n_{52}\beta_{52} : \varepsilon = \pm 1, m_{52}, n_{52} \in \mathbb{Z} \} \cong H_{52} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{52} \trianglelefteq G_{52}$ and the number of G_{52}/Γ_{52} -orbits of vertices of X_{52} is six. Therefore, G_{52}/Γ_{52} acts on $X_{52} = K_{52}/\Gamma_{52}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_1 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_3 \rangle$, $O_6 = \langle b_2 \rangle$ are the G_{52} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_1 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_3 \rangle$, $O_6 = \langle b_2 \rangle$ are the (G_{52}/Γ_{52}) -orbits. Since the vertex set of X_{52} is $\sqcup_{j=1}^6 \eta_{52}(O_j)$ and $G_{52}/\Gamma_{52} \leq \text{Aut}(X_{52})$, it follows that the number of $\text{Aut}(X_{52})$ -orbits of vertices is ≤ 6 .

Let $X_{53} = K_{53}/\Gamma_{53}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{53} \leq \text{Aut}(K_{53})$. Let the vertices of X_{53} form m_{53} $\text{Aut}(X_{53})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_1 as the origin $(0, 0)$ of K_{53} (see Section 4). Let $\alpha_{53} := a_1 - a_0$ and $\beta_{53} := a_{42} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{53} := \langle x \mapsto x + \alpha_{53}, x \mapsto x + \beta_{53} \rangle$ and

$$G_{53} = \{ \alpha : x \mapsto \varepsilon x + m_{53}\alpha_{53} + n_{53}\beta_{53} : \varepsilon = \pm 1, m_{53}, n_{53} \in \mathbb{Z} \} \cong H_{53} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{53} \trianglelefteq G_{53}$ and the number of G_{53}/Γ_{53} -orbits of vertices of X_{53} is five. Therefore, G_{53}/Γ_{53} acts on $X_{53} = K_{53}/\Gamma_{53}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{23} \rangle$, $O_5 = \langle a_{32} \rangle$ are the G_{53} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle b_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{23} \rangle$, $O_5 = \langle a_{32} \rangle$ are the (G_{53}/Γ_{53}) -orbits. Since the vertex set of X_{53} is $\sqcup_{j=1}^5 \eta_{53}(O_j)$ and $G_{53}/\Gamma_{53} \leq \text{Aut}(X_{53})$, it follows that the number of $\text{Aut}(X_{53})$ -orbits of vertices is ≤ 5 .

Let $X_{54} = K_{54}/\Gamma_{54}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{54} \leq \text{Aut}(K_{54})$. Let the vertices of X_{54} form m_{54} $\text{Aut}(X_{54})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_{14} as the origin $(0, 0)$ of K_{54} (see Section 4). Let $\alpha_{54} := a_1 - a_0$ and $\beta_{54} := a_{81} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{54} := \langle x \mapsto x + \alpha_{54}, x \mapsto x + \beta_{54} \rangle$ and

$$G_{54} = \{ \alpha : x \mapsto \varepsilon x + m_{54}\alpha_{54} + n_{54}\beta_{54} : \varepsilon = \pm 1, m_{54}, n_{54} \in \mathbb{Z} \} \cong H_{54} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{54} \trianglelefteq G_{54}$ and the number of G_{54}/Γ_{54} -orbits of vertices of X_{54} is three. Therefore, G_{54}/Γ_{54} acts on $X_{54} = K_{54}/\Gamma_{54}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_{51} \rangle$, $O_3 = \langle b_0 \rangle$, are the G_{54} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_{51} \rangle$, $O_3 = \langle b_0 \rangle$ are the (G_{54}/Γ_{54}) -orbits. Since the vertex set of X_{54} is $\sqcup_{j=1}^3 \eta_{54}(O_j)$ and $G_{54}/\Gamma_{54} \leq \text{Aut}(X_{54})$, it follows that the number of $\text{Aut}(X_{54})$ -orbits of vertices is 3.

Let $X_{55} = K_{55}/\Gamma_{55}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{55} \leq \text{Aut}(K_{55})$. Let the vertices of X_{55} form m_{55} $\text{Aut}(X_{55})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_1 as the origin $(0, 0)$ of K_{55} (see Section 4). Let $\alpha_{55} := a_1 - a_0$ and $\beta_{55} := a_{31} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{55} := \langle x \mapsto x + \alpha_{55}, x \mapsto x + \beta_{55} \rangle$ and

$$G_{55} = \{ \alpha : x \mapsto \varepsilon x + m_{55}\alpha_{55} + n_{55}\beta_{55} : \varepsilon = \pm 1, m_{55}, n_{55} \in \mathbb{Z} \} \cong H_{55} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{55} \trianglelefteq G_{55}$ and the number of G_{55}/Γ_{55} -orbits of vertices of X_{55} is five. Therefore, G_{55}/Γ_{55} acts on $X_{55} = K_{55}/\Gamma_{55}$. Since $O_1 = \langle b_0 \rangle$, $O_2 = \langle a_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{23} \rangle$, $O_5 = \langle a_{32} \rangle$ are the G_{55} -orbits, it follows that $O_1 = \langle b_0 \rangle$, $O_2 = \langle a_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{23} \rangle$, $O_5 = \langle a_{32} \rangle$ are the (G_{55}/Γ_{55}) -orbits. Since the vertex set of X_{55} is $\sqcup_{j=1}^5 \eta_{55}(O_j)$ and $G_{55}/\Gamma_{55} \leq \text{Aut}(X_{55})$, it follows that the number of $\text{Aut}(X_{55})$ -orbits of vertices is ≤ 5 .

Let $X_{56} = K_{56}/\Gamma_{56}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{56} \leq \text{Aut}(K_{56})$. Let the vertices of X_{56} form m_{56} $\text{Aut}(X_{56})$ -orbits. We take the middle point of the line segment joining vertices a_0 and b_1 as the origin $(0, 0)$ of K_{56} (see Section 4). Let $\alpha_{56} := a_1 - a_0$ and $\beta_{56} := a_{45} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{56} := \langle x \mapsto x + \alpha_{56}, x \mapsto x + \beta_{56} \rangle$ and

$$G_{56} = \{ \alpha : x \mapsto \varepsilon x + m_{56}\alpha_{56} + n_{56}\beta_{56} : \varepsilon = \pm 1, m_{56}, n_{56} \in \mathbb{Z} \} \cong H_{56} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{56} \trianglelefteq G_{56}$ and the number of G_{56}/Γ_{56} -orbits of vertices of X_{56} is six. Therefore, G_{56}/Γ_{56} acts on $X_{56} = K_{56}/\Gamma_{56}$. Since $O_1 = \langle b_0 \rangle$, $O_2 = \langle a_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{25} \rangle$, $O_5 = \langle a_{32} \rangle$, $O_6 = \langle a_{33} \rangle$ are the G_{56} -orbits, it follows that $O_1 = \langle b_0 \rangle$, $O_2 = \langle a_0 \rangle$, $O_3 = \langle a_{12} \rangle$, $O_4 = \langle a_{25} \rangle$, $O_5 = \langle a_{32} \rangle$, $O_6 = \langle a_{33} \rangle$ are the (G_{56}/Γ_{56}) -orbits. Since the vertex set of X_{56} is $\sqcup_{j=1}^6 \eta_{56}(O_j)$ and $G_{56}/\Gamma_{56} \leq \text{Aut}(X_{56})$, it follows that the number of $\text{Aut}(X_{56})$ -orbits of vertices is ≤ 6 .

Let $X_{57} = K_{57}/\Gamma_{57}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{57} \leq \text{Aut}(K_{57})$. Let the vertices of X_{57} form m_{57} $\text{Aut}(X_{57})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_1 as the origin $(0, 0)$ of K_{57} (see Section 4). Let $\alpha_{57} := b_1 - b_0$ and $\beta_{57} := b_{20} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{57} := \langle x \mapsto x + \alpha_{57}, x \mapsto x + \beta_{57} \rangle$ and

$$G_{57} = \{ \alpha : x \mapsto \varepsilon x + m_{57}\alpha_{57} + n_{57}\beta_{57} : \varepsilon = \pm 1, m_{57}, n_{57} \in \mathbb{Z} \} \cong H_{57} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{57} \trianglelefteq G_{57}$ and the number of G_{57}/Γ_{57} -orbits of vertices of X_{57} is six. Therefore, G_{57}/Γ_{57} acts on $X_{57} = K_{57}/\Gamma_{57}$. Since $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_0 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_9 \rangle$, $O_5 = \langle b_8 \rangle$, $O_6 = \langle b_{15} \rangle$ are the G_{57} -orbits, it follows that $O_1 = \langle a_1 \rangle$, $O_2 = \langle a_0 \rangle$, $O_3 = \langle b_0 \rangle$, $O_4 = \langle b_9 \rangle$, $O_5 = \langle b_8 \rangle$, $O_6 = \langle b_{15} \rangle$ are the (G_{57}/Γ_{57}) -orbits. Since the vertex set of X_{56} is $\sqcup_{j=1}^6 \eta_{57}(O_j)$ and $G_{57}/\Gamma_{57} \leq \text{Aut}(X_{57})$, it follows that the number of $\text{Aut}(X_{57})$ -orbits of vertices is ≤ 6 .

Let $X_{58} = K_{58}/\Gamma_{58}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{58} \leq \text{Aut}(K_{58})$. Let the vertices of X_{58} form m_{58} $\text{Aut}(X_{58})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin

$(0,0)$ of K_{58} (see Section 4). Let $\alpha_{58} := a_{24} - a_0$, $\beta_{58} := a_{34} - a_0$ and $\gamma_{58} := a_{56} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{58} := \langle x \mapsto x + \alpha_{58}, x \mapsto x + \beta_{58}, x \mapsto x + \gamma_{58} \rangle$ and

$$G_{58} = \{\alpha : x \mapsto \varepsilon x + m_{58}\alpha_{58} + n_{58}\beta_{58} + r_{58}\gamma_{58} : \varepsilon = \pm 1, m_{58}, n_{58}, r_{58} \in \mathbb{Z}\} \cong H_{58} \rtimes \mathbb{Z}_2$$

By the same arguments as above and in Claim 1, $\Gamma_{58} \trianglelefteq G_{58}$ and the number of G_{58}/Γ_{58} -orbits of vertices of X_{58} is twelve. Therefore, G_{58}/Γ_{58} acts on $X_{58} = K_{58}/\Gamma_{58}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle a_6 \rangle$, $O_6 = \langle b_3 \rangle$, $O_7 = \langle b_4 \rangle$, $O_8 = \langle a_8 \rangle$, $O_9 = \langle b_7 \rangle$, $O_{10} = \langle b_8 \rangle$, $O_{11} = \langle b_9 \rangle$, $O_{12} = \langle a_{10} \rangle$ are the G_{58} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_0 \rangle$, $O_5 = \langle a_6 \rangle$, $O_6 = \langle b_3 \rangle$, $O_7 = \langle b_4 \rangle$, $O_8 = \langle a_8 \rangle$, $O_9 = \langle b_7 \rangle$, $O_{10} = \langle b_8 \rangle$, $O_{11} = \langle b_9 \rangle$, $O_{12} = \langle a_{10} \rangle$ are the (G_{58}/Γ_{58}) -orbits. Since the vertex set of X_{58} is $\sqcup_{j=1}^{12} \eta_{58}(O_j)$ and $G_{58}/\Gamma_{58} \leq \text{Aut}(X_{58})$, it follows that the number of $\text{Aut}(X_{58})$ -orbits of vertices is ≤ 12 .

Let $X_{59} = K_{59}/\Gamma_{59}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{59} \leq \text{Aut}(K_{59})$. Let the vertices of X_{59} form m_{59} $\text{Aut}(X_{59})$ -orbits. We take the middle point of the line segment joining vertices b_0 and b_1 as the origin $(0,0)$ of K_{59} (see Section 4). Let $\alpha_{59} := b_1 - b_0$ and $\beta_{59} := b_{18} - b_0 \in \mathbb{R}^2$. Similarly as above, define $H_{59} := \langle x \mapsto x + \alpha_{59}, x \mapsto x + \beta_{59} \rangle$ and

$$G_{59} = \{\alpha : x \mapsto \varepsilon x + m_{59}\alpha_{59} + n_{59}\beta_{59} : \varepsilon = \pm 1, m_{59}, n_{59} \in \mathbb{Z}\} \cong H_{59} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{59} \trianglelefteq G_{59}$ and the number of G_{59}/Γ_{59} -orbits of vertices of X_{59} is six. Therefore, G_{59}/Γ_{59} acts on $X_{59} = K_{59}/\Gamma_{59}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle a_9 \rangle$, $O_5 = \langle b_4 \rangle$, $O_6 = \langle a_{11} \rangle$ are the G_{59} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle b_1 \rangle$, $O_4 = \langle a_9 \rangle$, $O_5 = \langle b_4 \rangle$, $O_6 = \langle a_{11} \rangle$ are the (G_{59}/Γ_{59}) -orbits. Since the vertex set of X_{59} is $\sqcup_{j=1}^6 \eta_{59}(O_j)$ and $G_{59}/\Gamma_{59} \leq \text{Aut}(X_{59})$, it follows that the number of $\text{Aut}(X_{59})$ -orbits of vertices is ≤ 6 .

Let $X_{60} = K_{60}/\Gamma_{60}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{60} \leq \text{Aut}(K_{60})$. Let the vertices of X_{60} form m_{60} $\text{Aut}(X_{60})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{60} (see Section 4). Let $\alpha_{60} := a_3 - a_0$ and $\beta_{60} := a_{54} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{60} := \langle x \mapsto x + \alpha_{60}, x \mapsto x + \beta_{60} \rangle$ and

$$G_{60} = \{\alpha : x \mapsto \varepsilon x + m_{60}\alpha_{60} + n_{60}\beta_{60} : \varepsilon = \pm 1, m_{60}, n_{60} \in \mathbb{Z}\} \cong H_{60} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{60} \trianglelefteq G_{60}$ and the number of G_{60}/Γ_{60} -orbits of vertices of X_{60} is eight. Therefore, G_{60}/Γ_{60} acts on $X_{60} = K_{60}/\Gamma_{60}$. Since $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_{15} \rangle$, $O_7 = \langle b_{14} \rangle$, $O_8 = \langle a_{15} \rangle$ are the G_{60} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle b_1 \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_{15} \rangle$, $O_7 = \langle b_{14} \rangle$, $O_8 = \langle a_{15} \rangle$ are the (G_{60}/Γ_{60}) -orbits. Since the vertex set of X_{60} is $\sqcup_{j=1}^8 \eta_{60}(O_j)$ and $G_{60}/\Gamma_{60} \leq \text{Aut}(X_{60})$, it follows that the number of $\text{Aut}(X_{60})$ -orbits of vertices is ≤ 8 .

Let $X_{61} = K_{61}/\Gamma_{61}$ be a semiequivelar map on the torus for some fixed element (vertex, edge or face) free subgroup $\Gamma_{61} \leq \text{Aut}(K_{61})$. Let the vertices of X_{61} form m_{61} $\text{Aut}(X_{61})$ -orbits. We take the middle point of the line segment joining vertices a_0 and a_3 as the origin $(0,0)$ of K_{61} (see Section 4). Let $\alpha_{61} := a_3 - a_0$ and $\beta_{61} := a_{49} - a_0 \in \mathbb{R}^2$. Similarly as above, define $H_{61} := \langle x \mapsto x + \alpha_{61}, x \mapsto x + \beta_{61} \rangle$ and

$$G_{61} = \{\alpha : x \mapsto \varepsilon x + m_{61}\alpha_{61} + n_{61}\beta_{61} : \varepsilon = \pm 1, m_{61}, n_{61} \in \mathbb{Z}\} \cong H_{61} \rtimes \mathbb{Z}_2.$$

By the same arguments as above and in Claim 1, $\Gamma_{61} \trianglelefteq G_{61}$ and the number of G_{61}/Γ_{61} -orbits of vertices of X_{61} is eight. Therefore, G_{61}/Γ_{61} acts on $X_{61} = K_{61}/\Gamma_{61}$. Since $O_1 = \langle a_0 \rangle$,

$O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_{15} \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_1 \rangle$, $O_6 = \langle b_{14} \rangle$, $O_6 = \langle b_{15} \rangle$ are the G_{61} -orbits, it follows that $O_1 = \langle a_0 \rangle$, $O_2 = \langle a_1 \rangle$, $O_3 = \langle a_2 \rangle$, $O_4 = \langle a_{15} \rangle$, $O_5 = \langle b_0 \rangle$, $O_6 = \langle b_1 \rangle$, $O_6 = \langle b_{14} \rangle$, $O_6 = \langle b_{15} \rangle$ are the (G_{61}/Γ_{61}) -orbits. Since the vertex set of X_{61} is $\sqcup_{j=1}^8 \eta_{61}(O_j)$ and $G_{61}/\Gamma_{61} \leq \text{Aut}(X_{61})$, it follows that the number of $\text{Aut}(X_{61})$ -orbits of vertices is ≤ 8 . $\square$

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7 Conflicts of Interest Statement

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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